

BFS Traversal

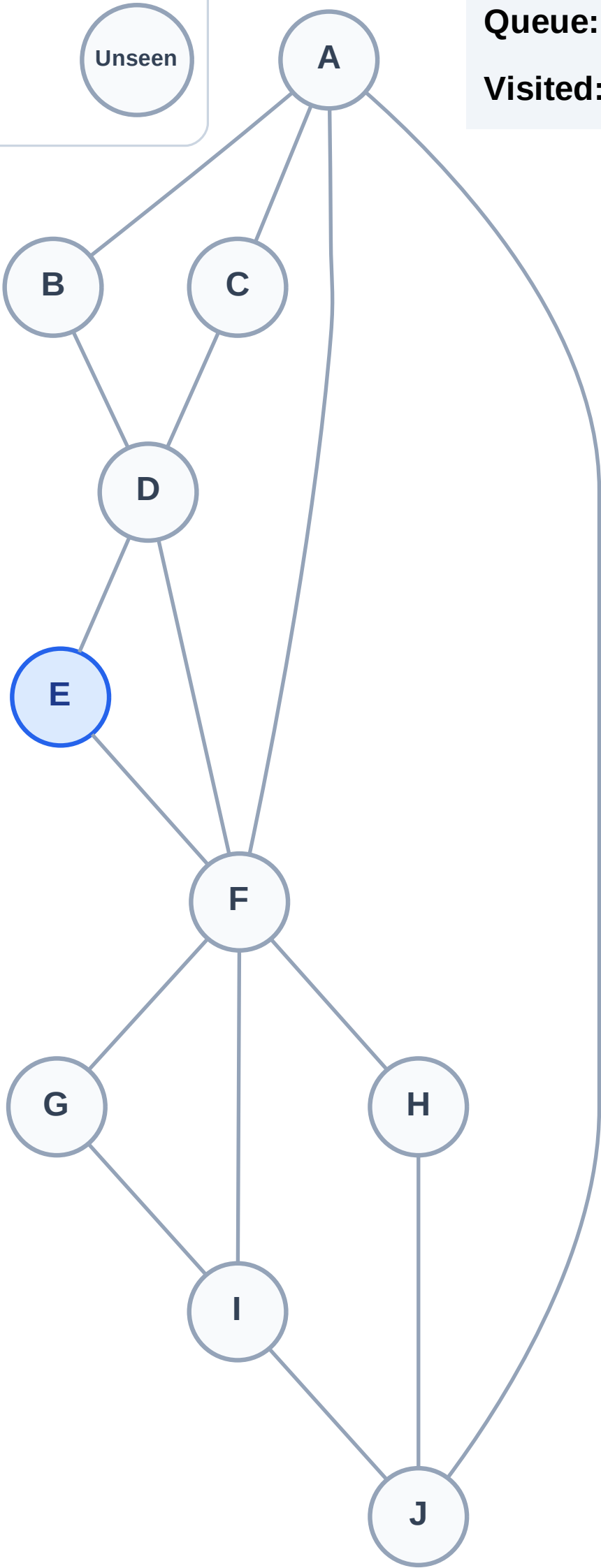
Step 1: Start at E

Legend



Queue: E

Visited: (none)



BFS Traversal

Step 2: Process node E

Legend

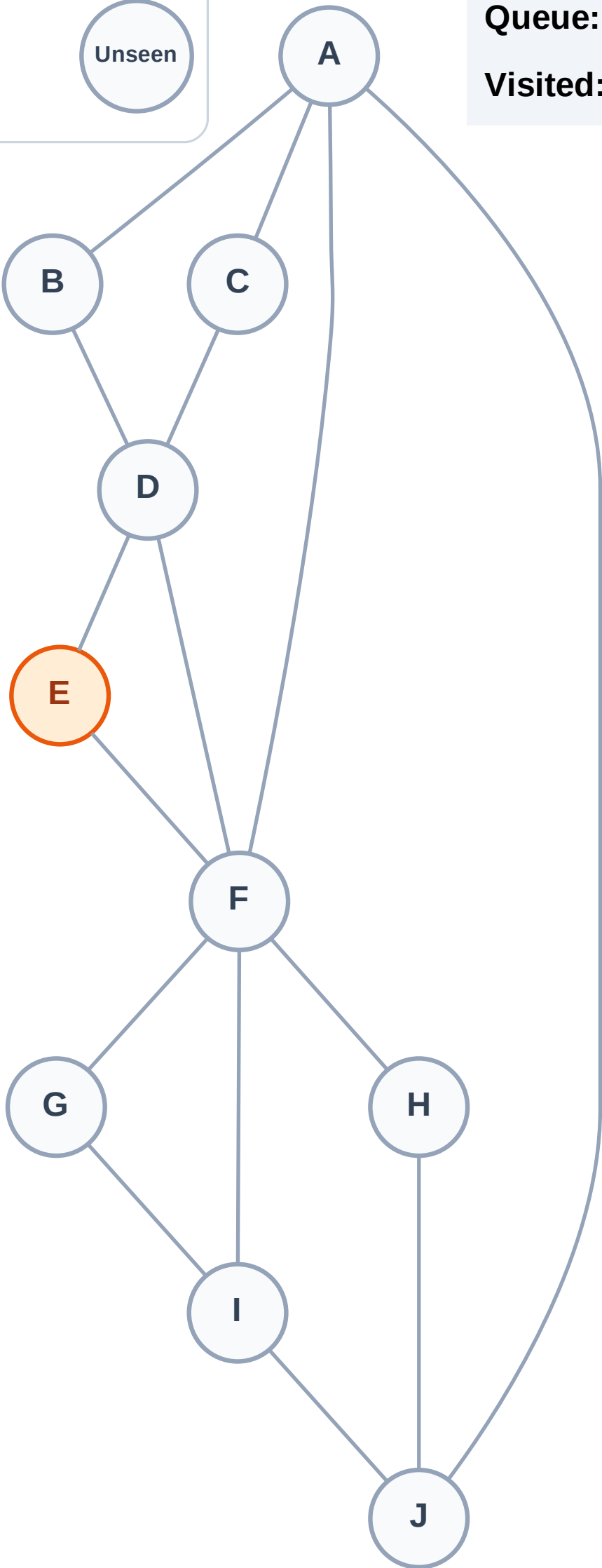
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

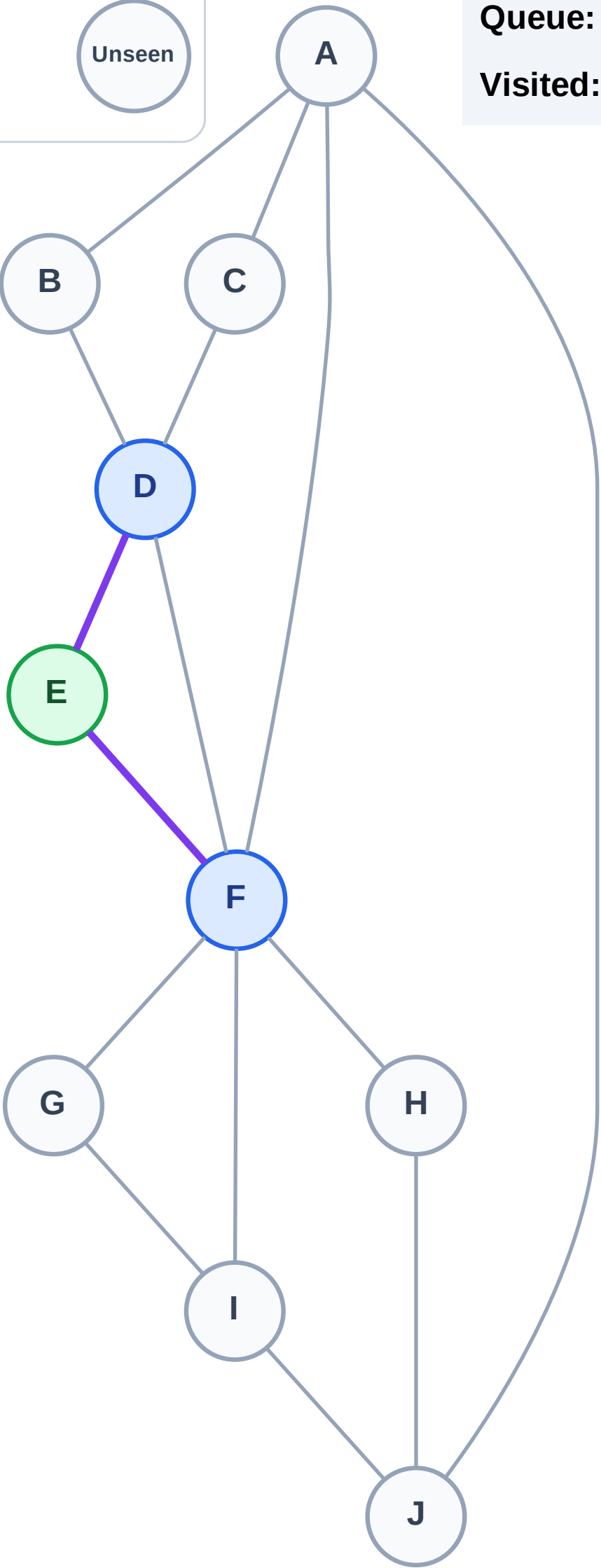
Step 3: Discover D, F from E

Legend



Queue: D → F

Visited: E



BFS Traversal

Step 4: Process node D

Legend

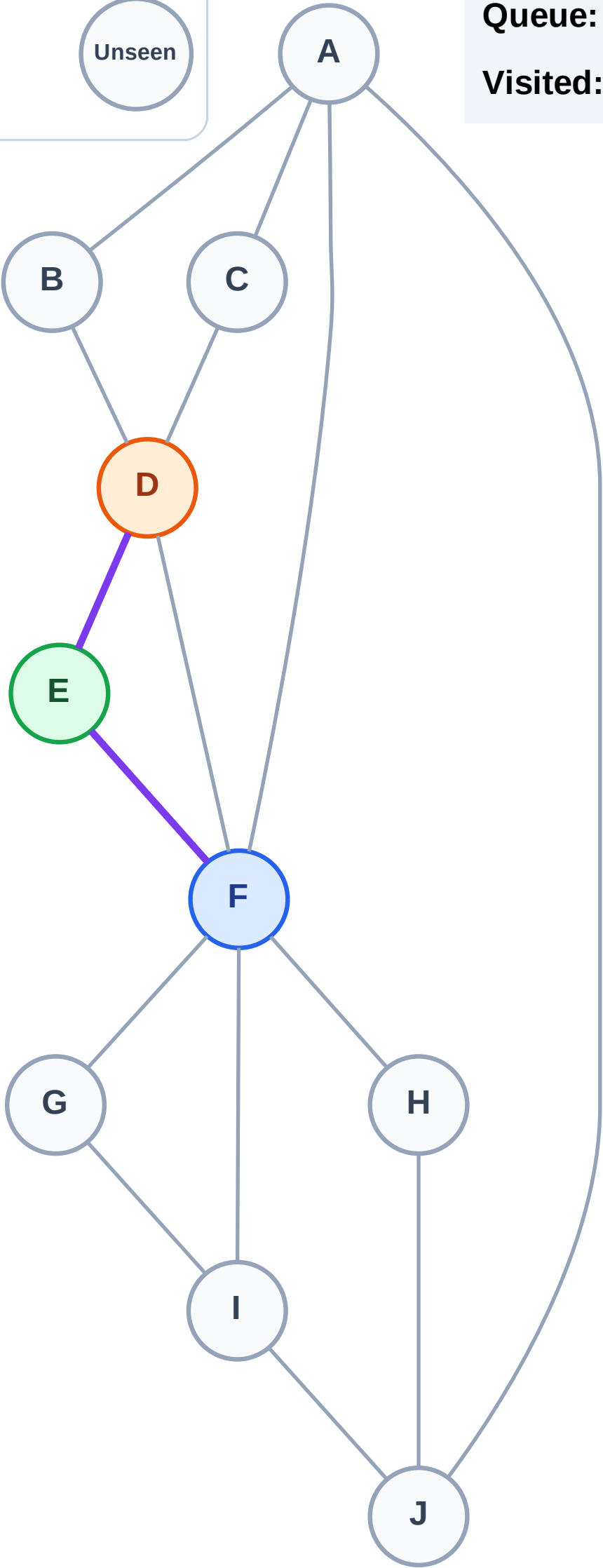
Complete

Processing

Discovered

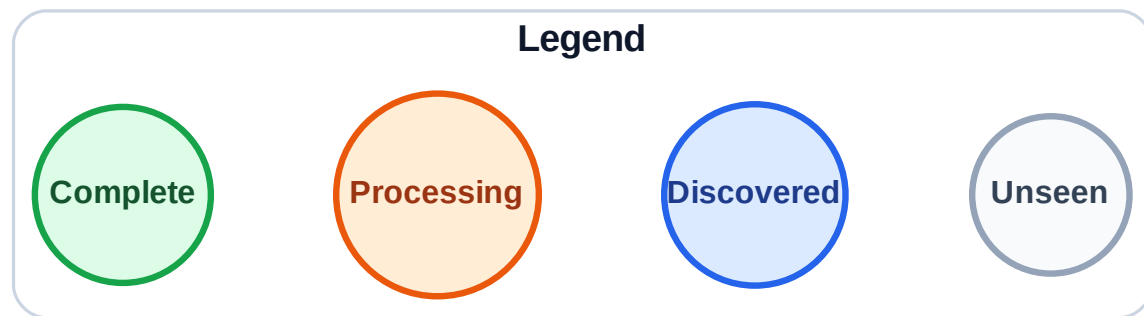
Unseen

Queue: F
Visited: E

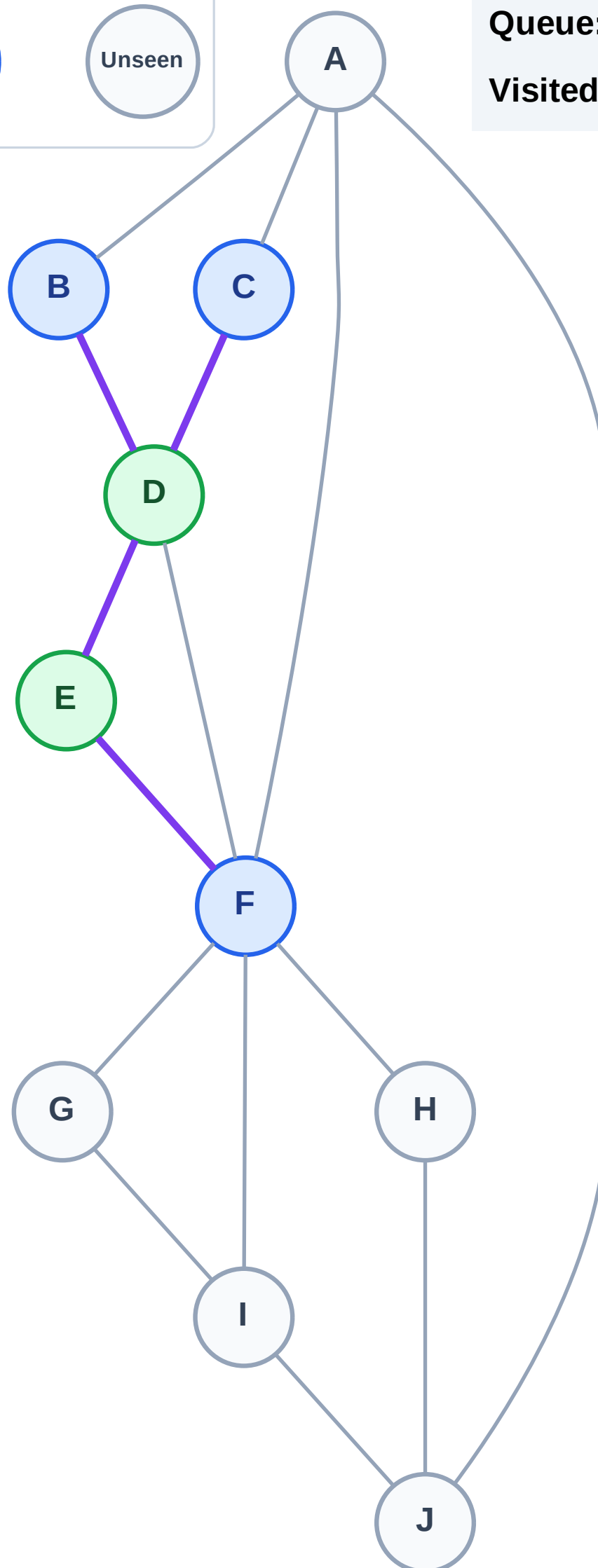


BFS Traversal

Step 5: Discover B, C from D



Queue: F → B → C
Visited: D, E



BFS Traversal

Step 6: Process node F

Legend

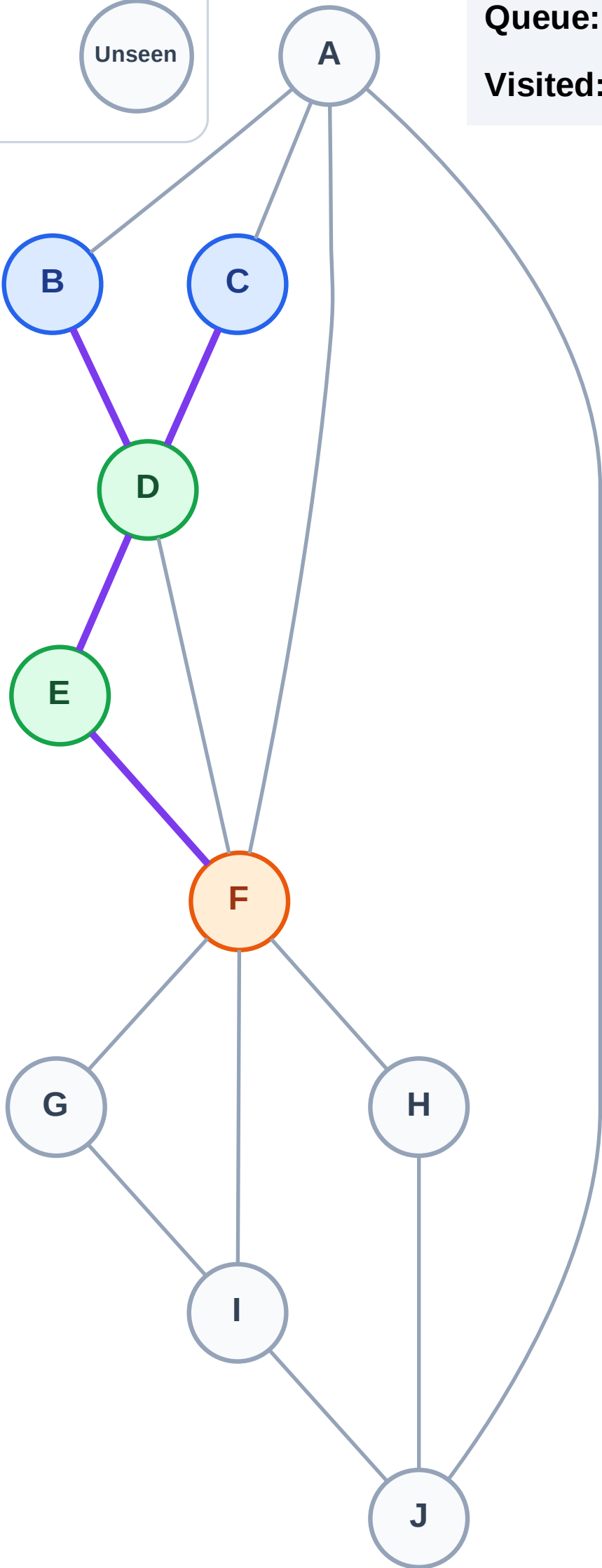
Complete

Processing

Discovered

Unseen

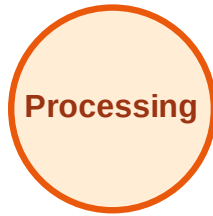
Queue: B → C
Visited: D, E



BFS Traversal

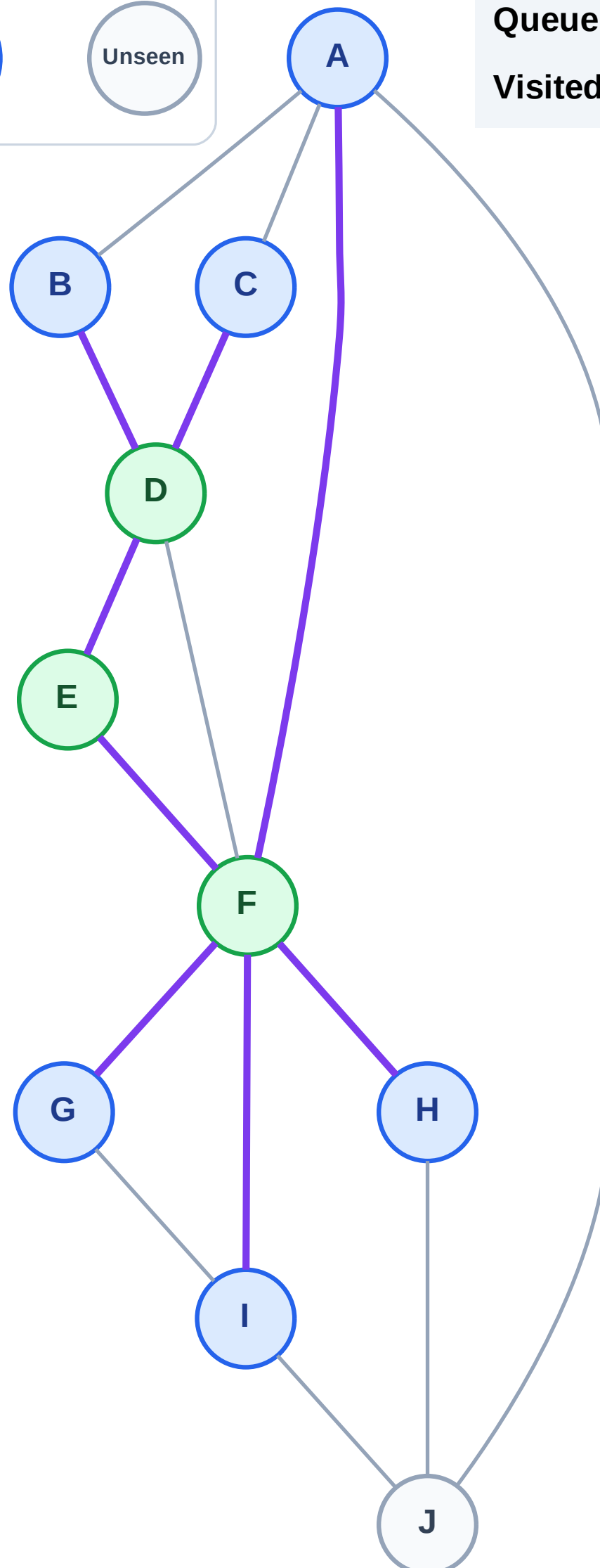
Step 7: Discover A, G, H, I from F

Legend



Queue: B → C → A → G → H → I

Visited: D, E, F



BFS Traversal

Step 8: Process node B

Legend

Complete

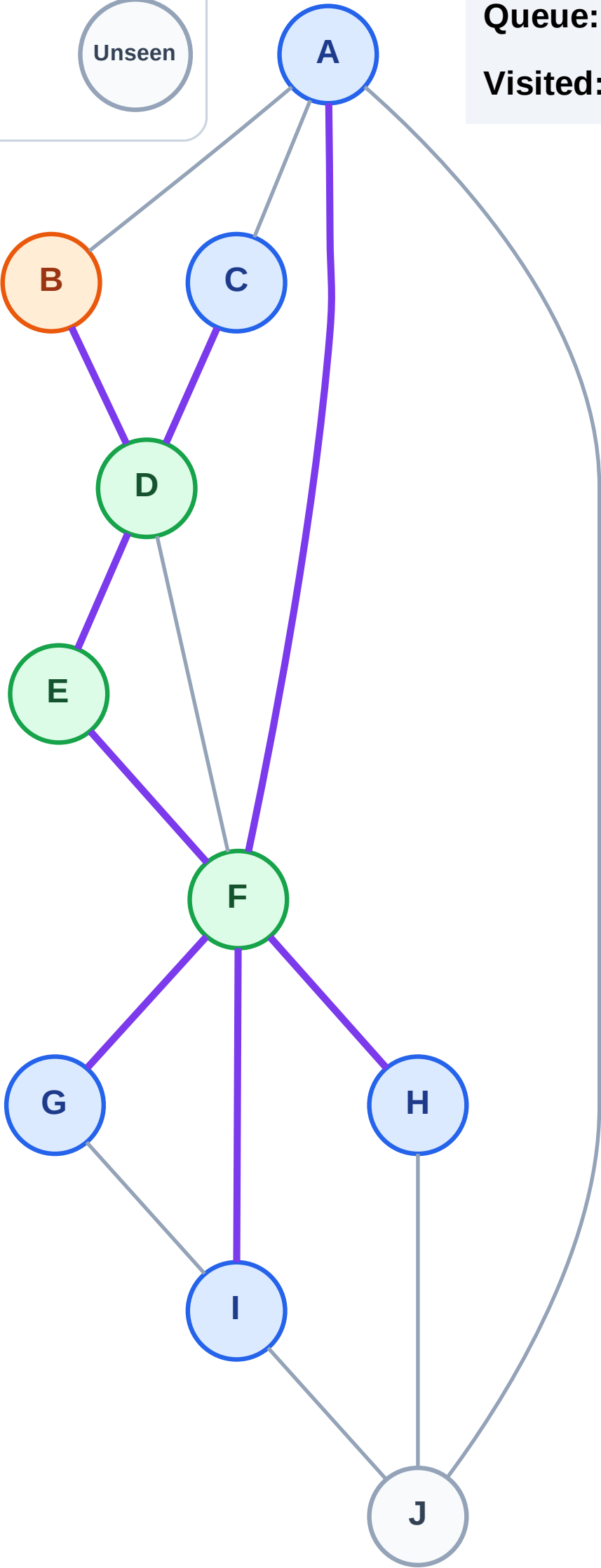
Processing

Discovered

Unseen

Queue: C → A → G → H → I

Visited: D, E, F



BFS Traversal

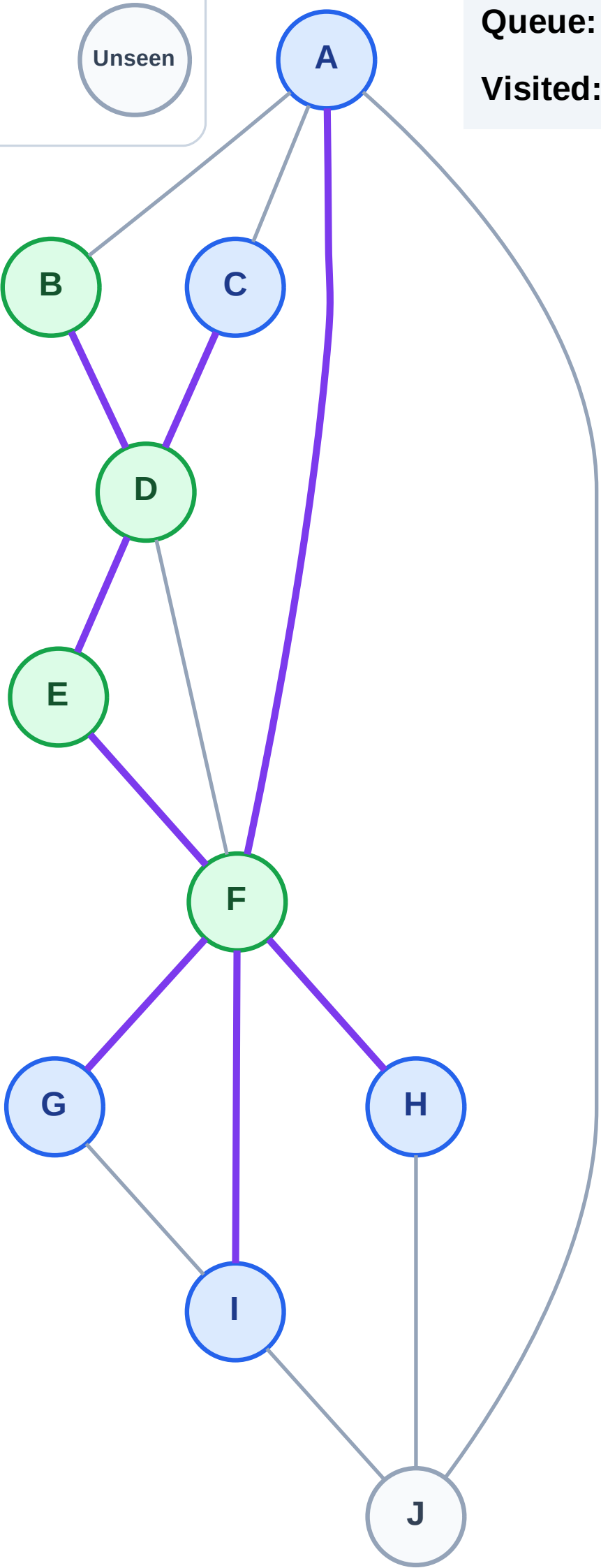
Step 9: No new neighbors from B

Legend



Queue: C → A → G → H → I

Visited: B, D, E, F



BFS Traversal

Step 10: Process node C

Legend

Complete

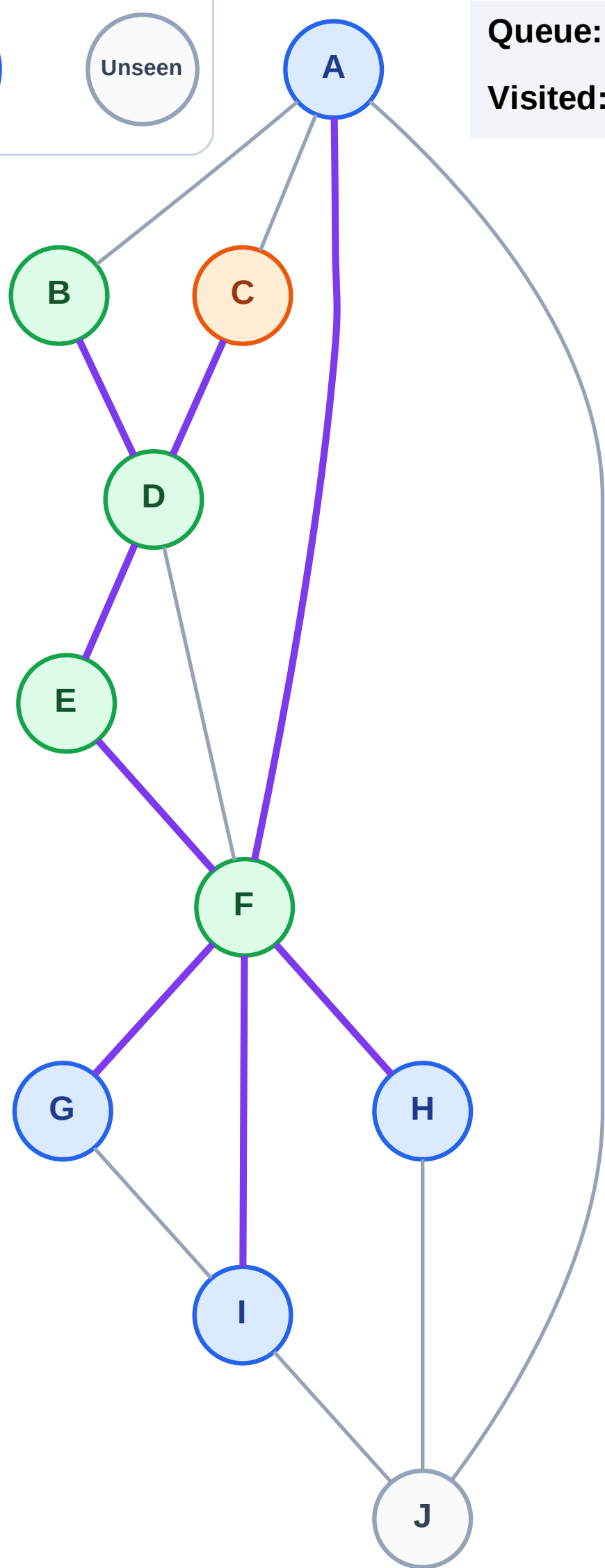
Processing

Discovered

Unseen

Queue: A → G → H → I

Visited: B, D, E, F



BFS Traversal

Step 11: No new neighbors from C

Legend

Complete

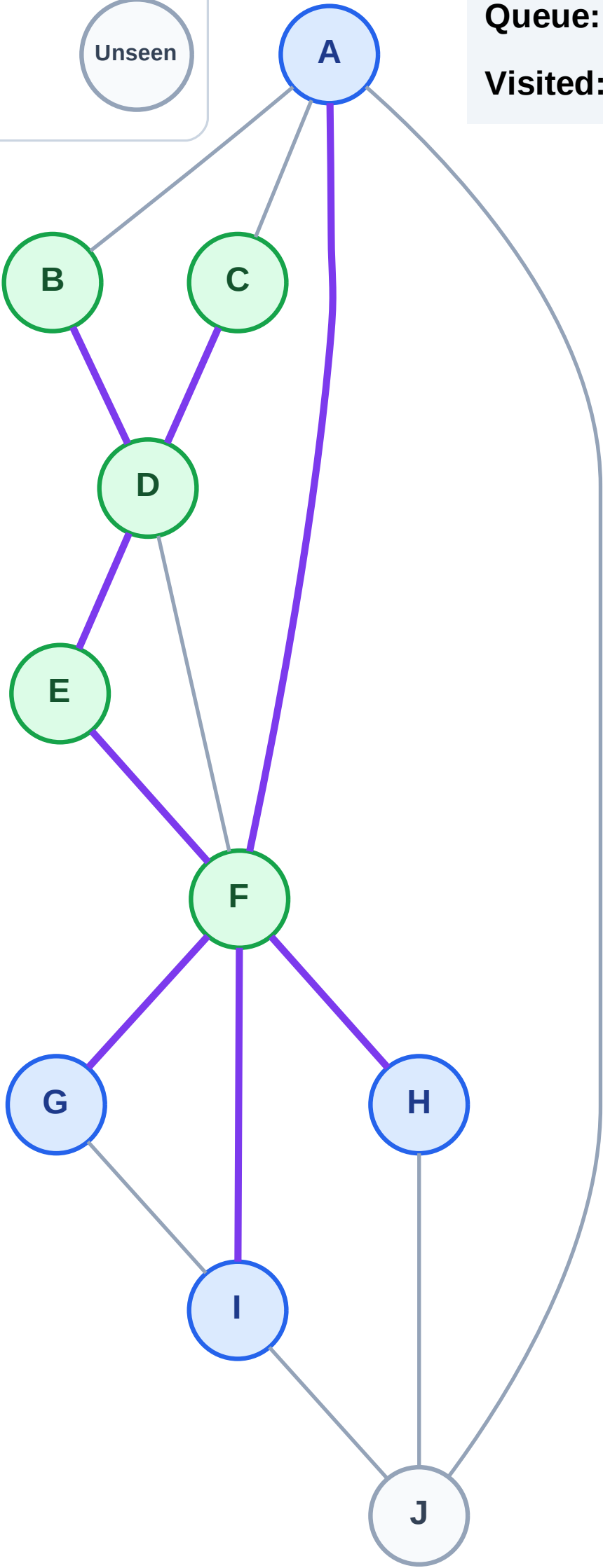
Processing

Discovered

Unseen

Queue: A → G → H → I

Visited: B, C, D, E, F



BFS Traversal

Step 12: Process node A

Legend

Complete

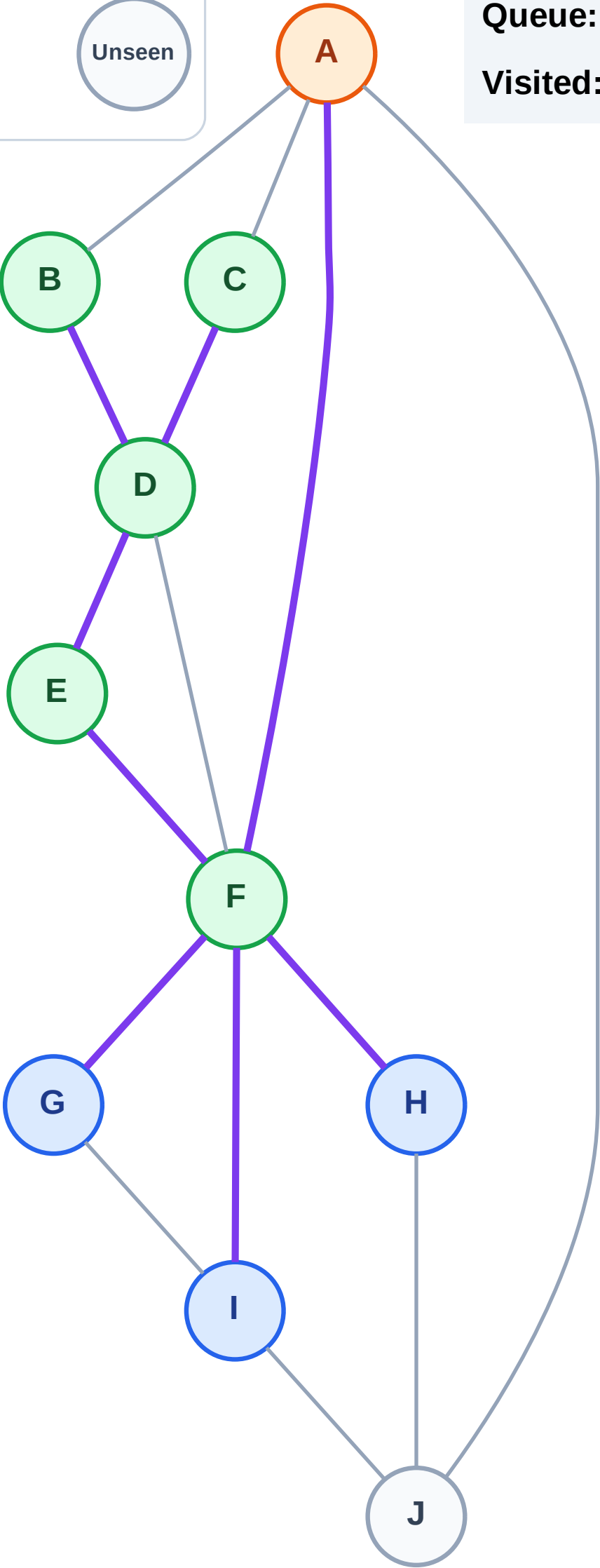
Processing

Discovered

Unseen

Queue: G → H → I

Visited: B, C, D, E, F



BFS Traversal

Step 13: Discover J from A

Legend

Complete

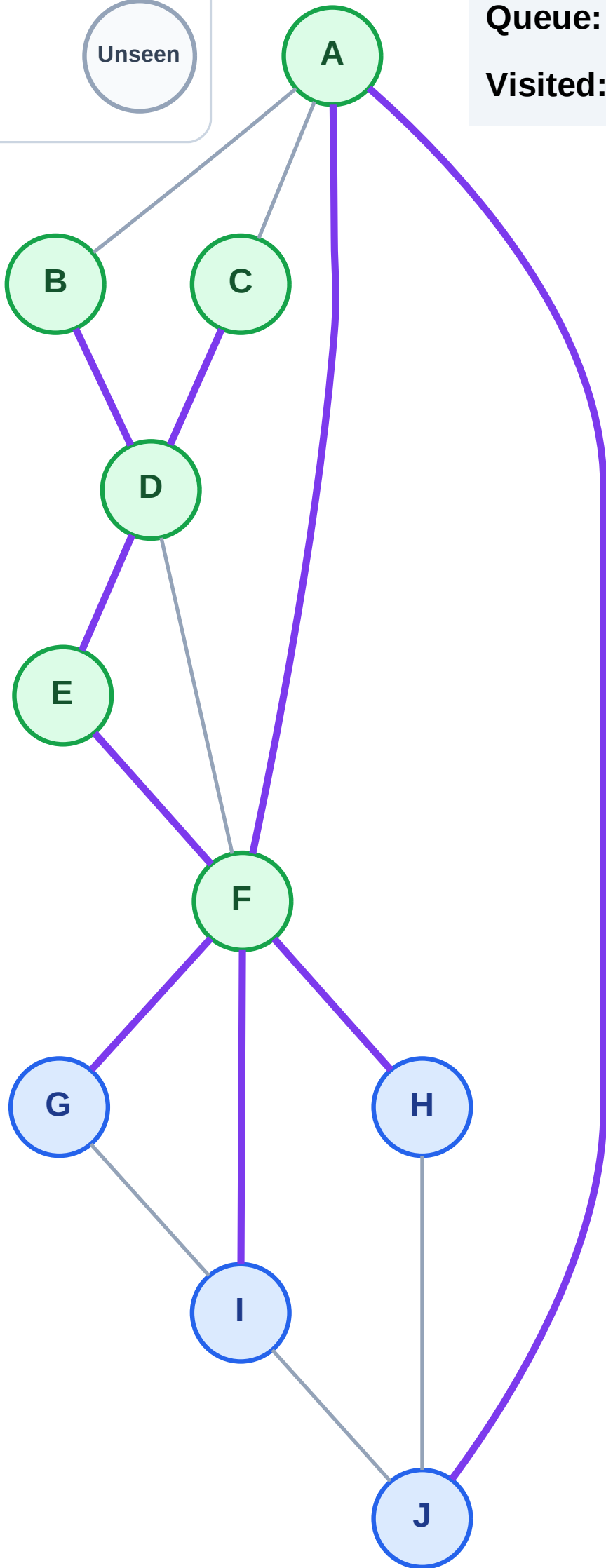
Processing

Discovered

Unseen

Queue: G → H → I → J

Visited: A, B, C, D, E, F



BFS Traversal

Step 14: Process node G

Legend

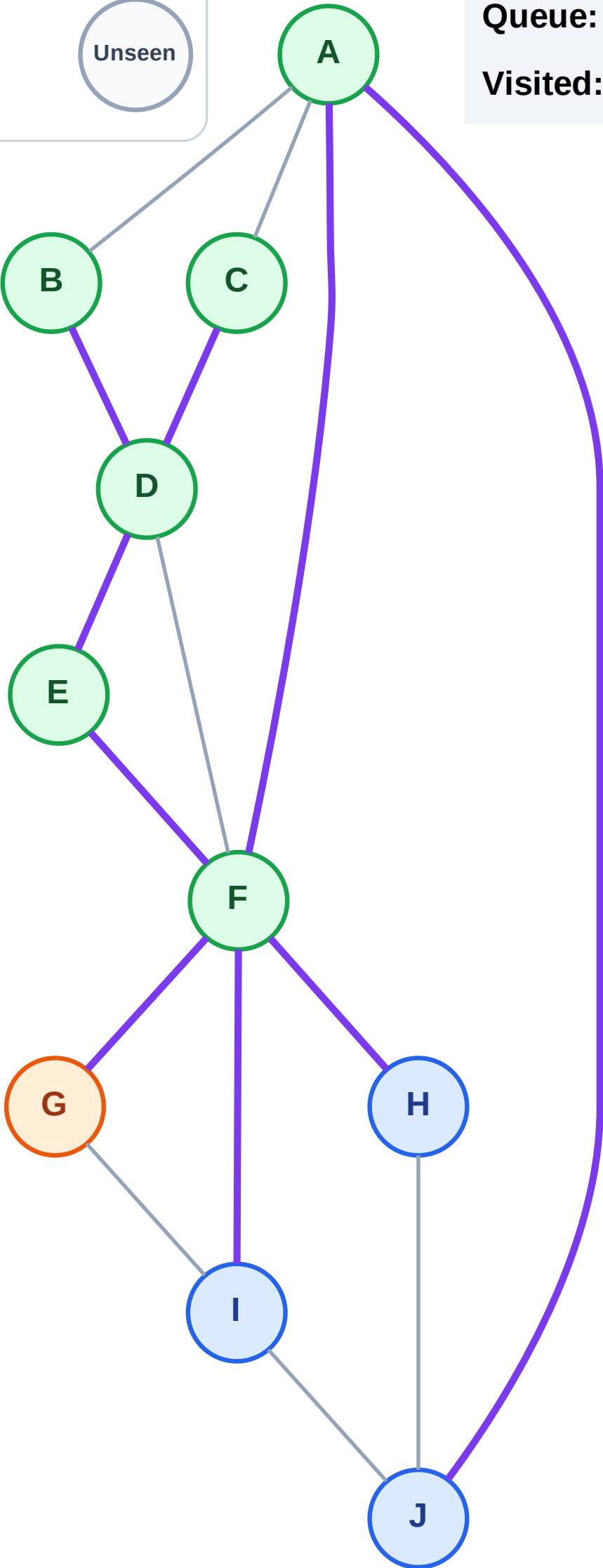
Complete

Processing

Discovered

Unseen

Queue: H → I → J
Visited: A, B, C, D, E, F



BFS Traversal

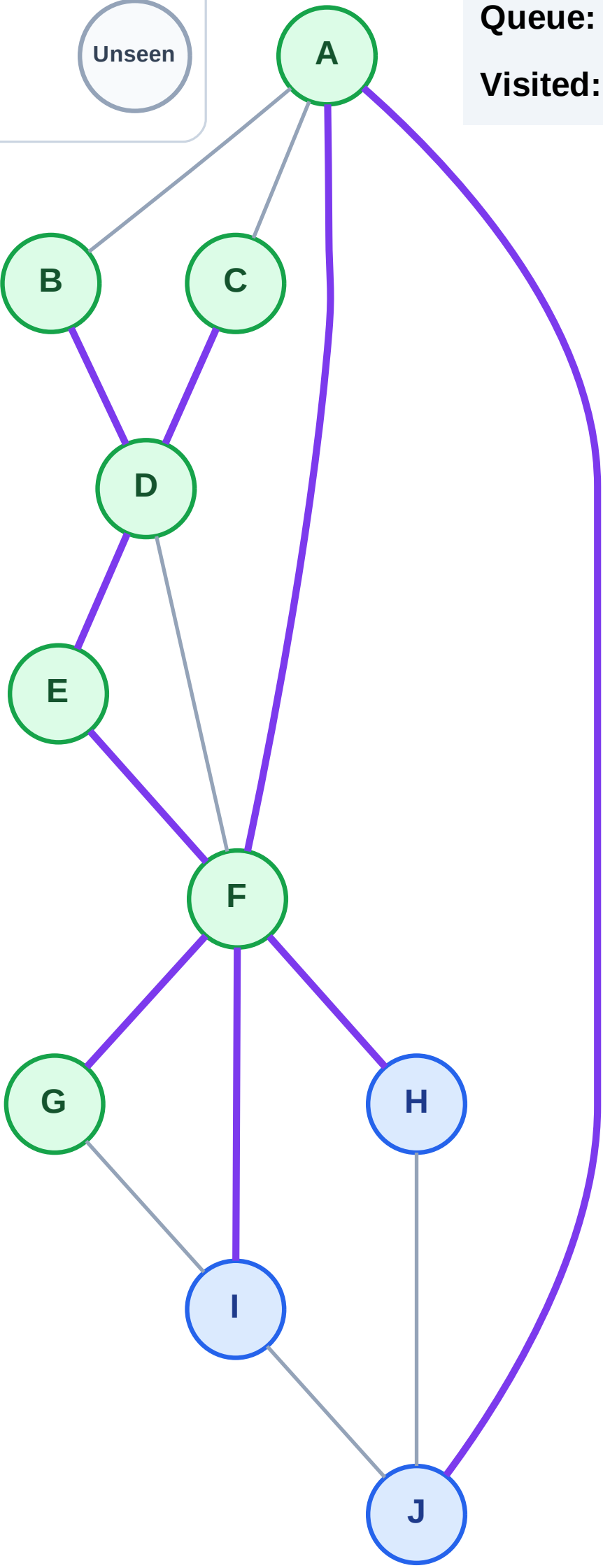
Step 15: No new neighbors from G

Legend



Queue: H → I → J

Visited: A, B, C, D, E, F, G



BFS Traversal

Step 16: Process node H

Legend

Complete

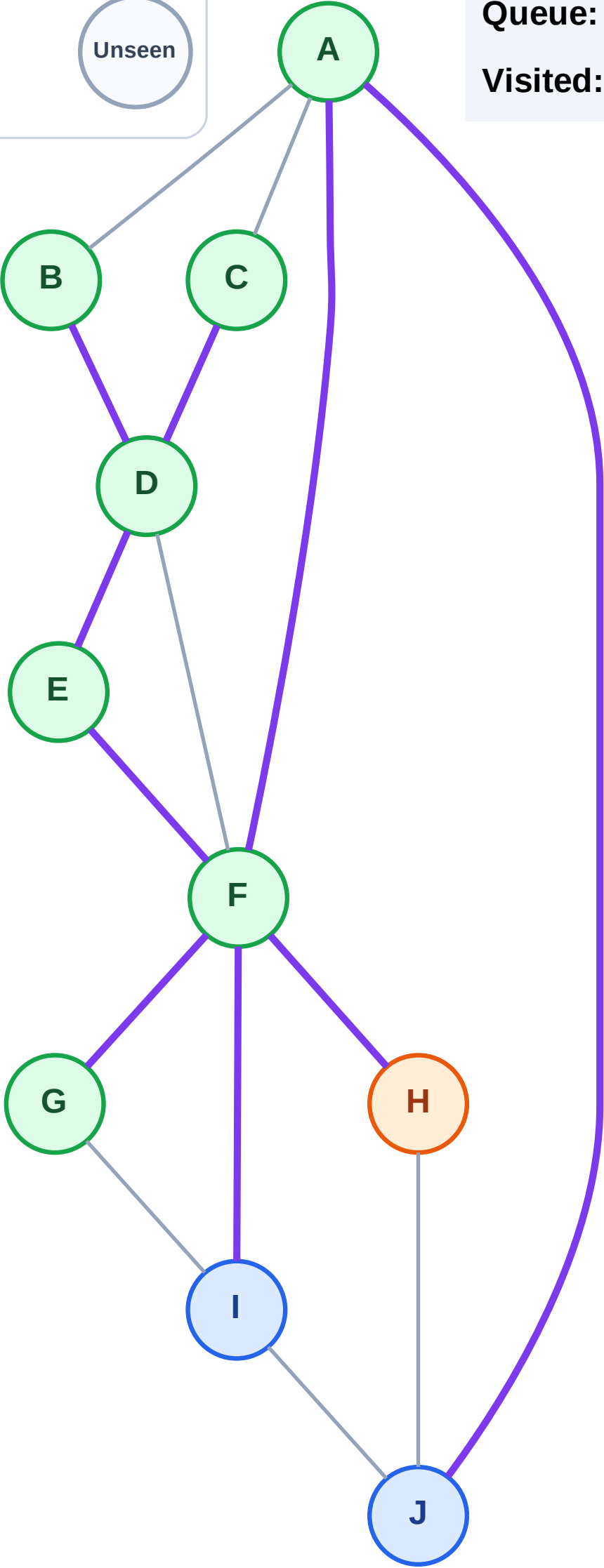
Processing

Discovered

Unseen

Queue: I → J

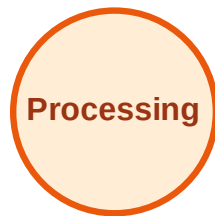
Visited: A, B, C, D, E, F, G



BFS Traversal

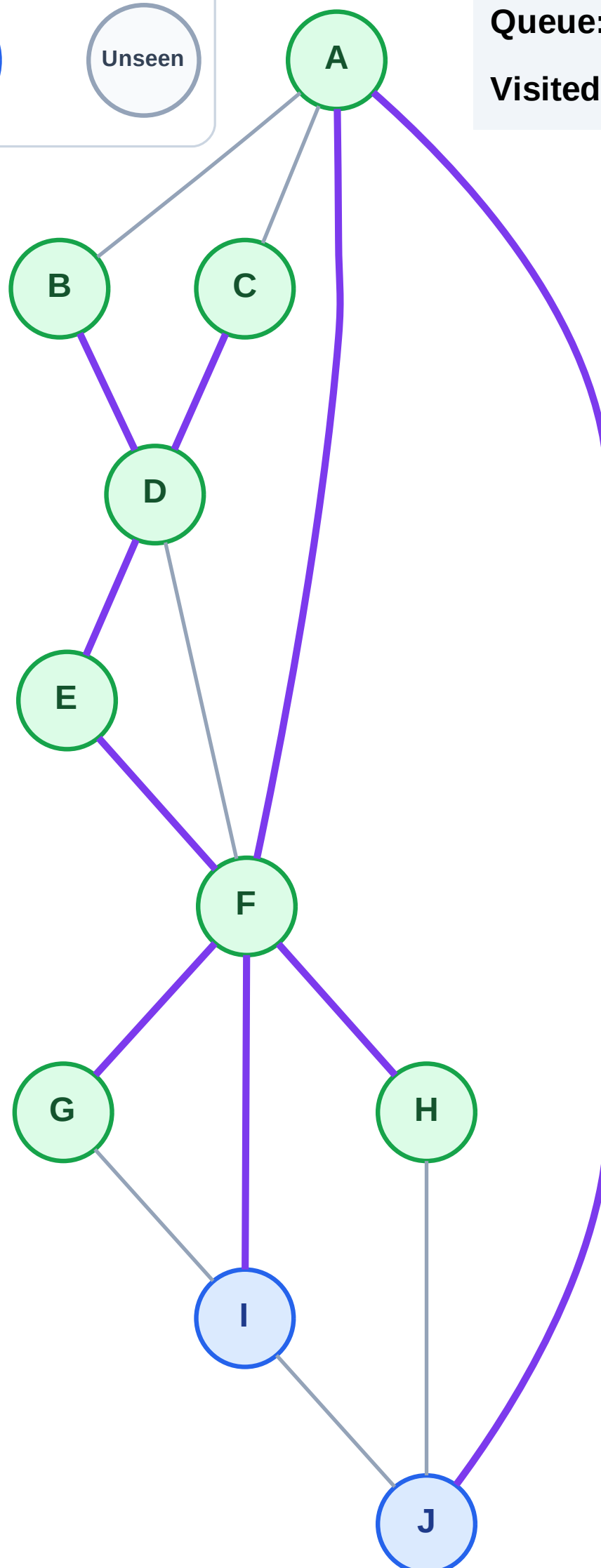
Step 17: No new neighbors from H

Legend



Queue: I → J

Visited: A, B, C, D, E, F, G, H



BFS Traversal

Step 18: Process node I

Legend

Complete

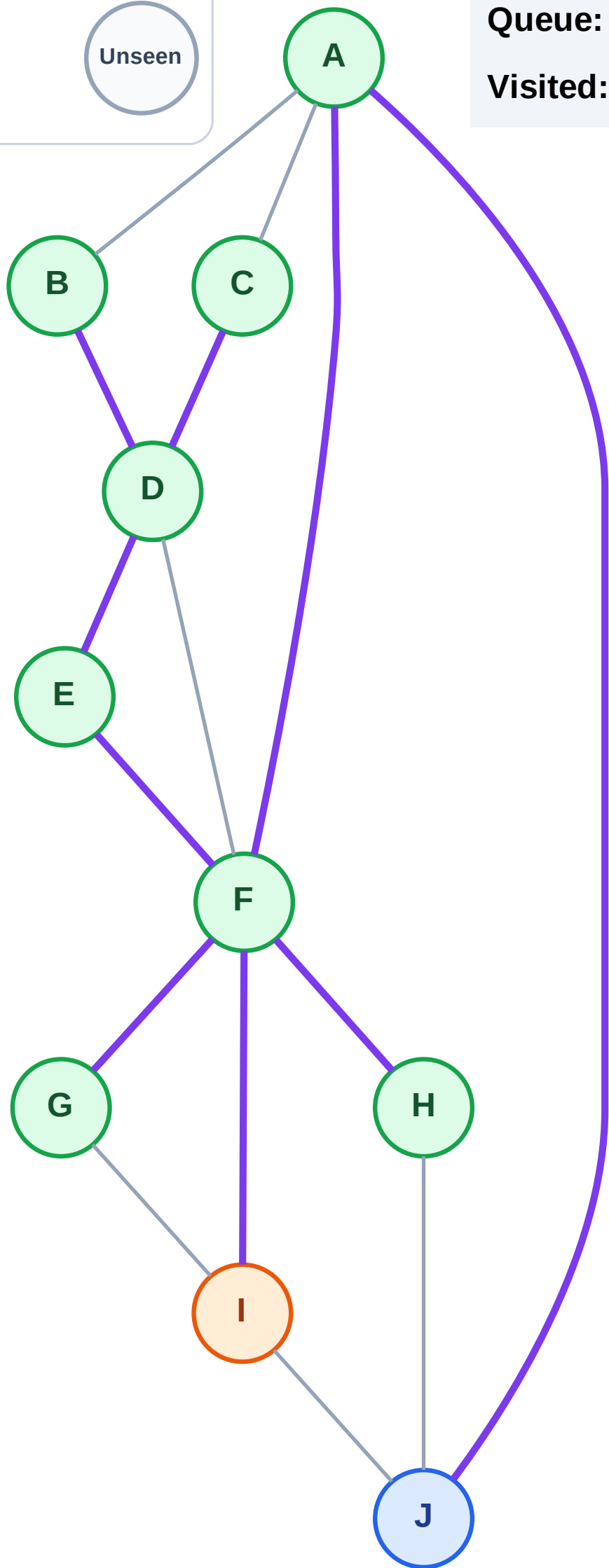
Processing

Discovered

Unseen

Queue: J

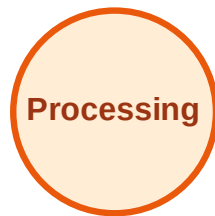
Visited: A, B, C, D, E, F, G, H



BFS Traversal

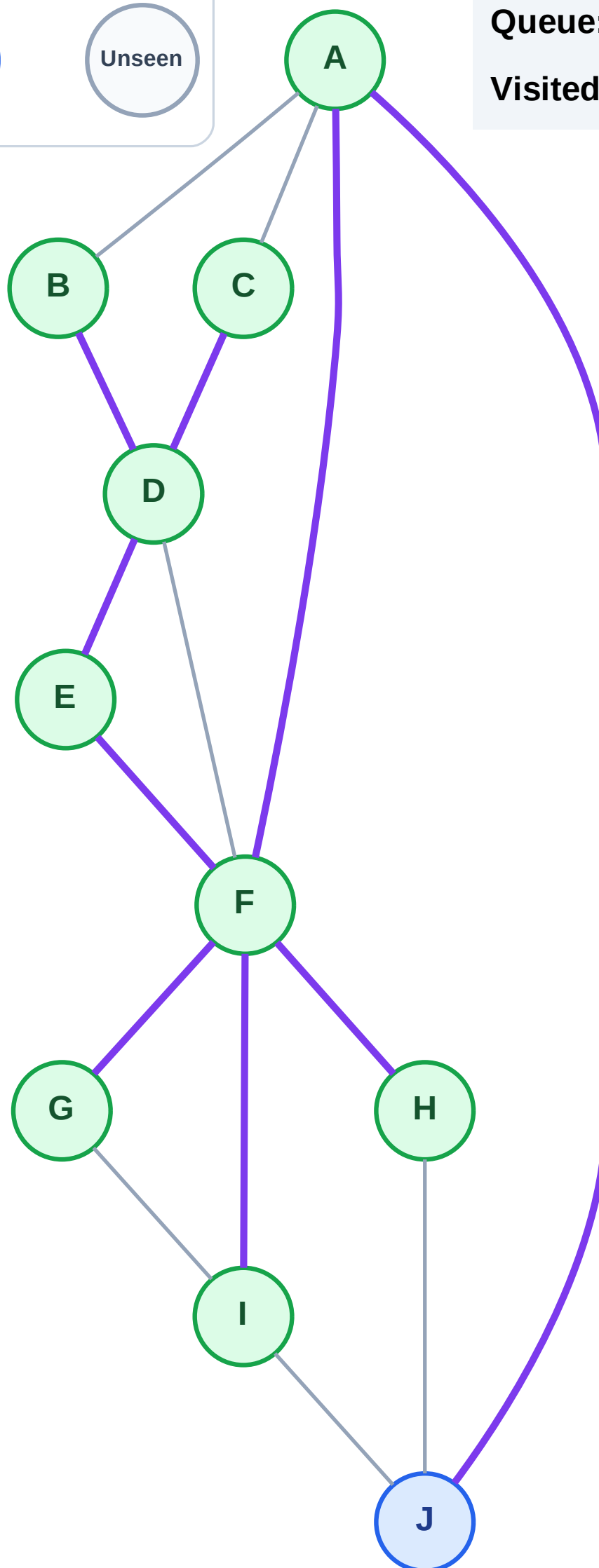
Step 19: No new neighbors from I

Legend



Queue: J

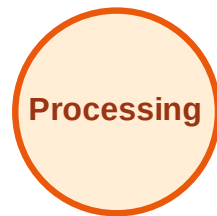
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

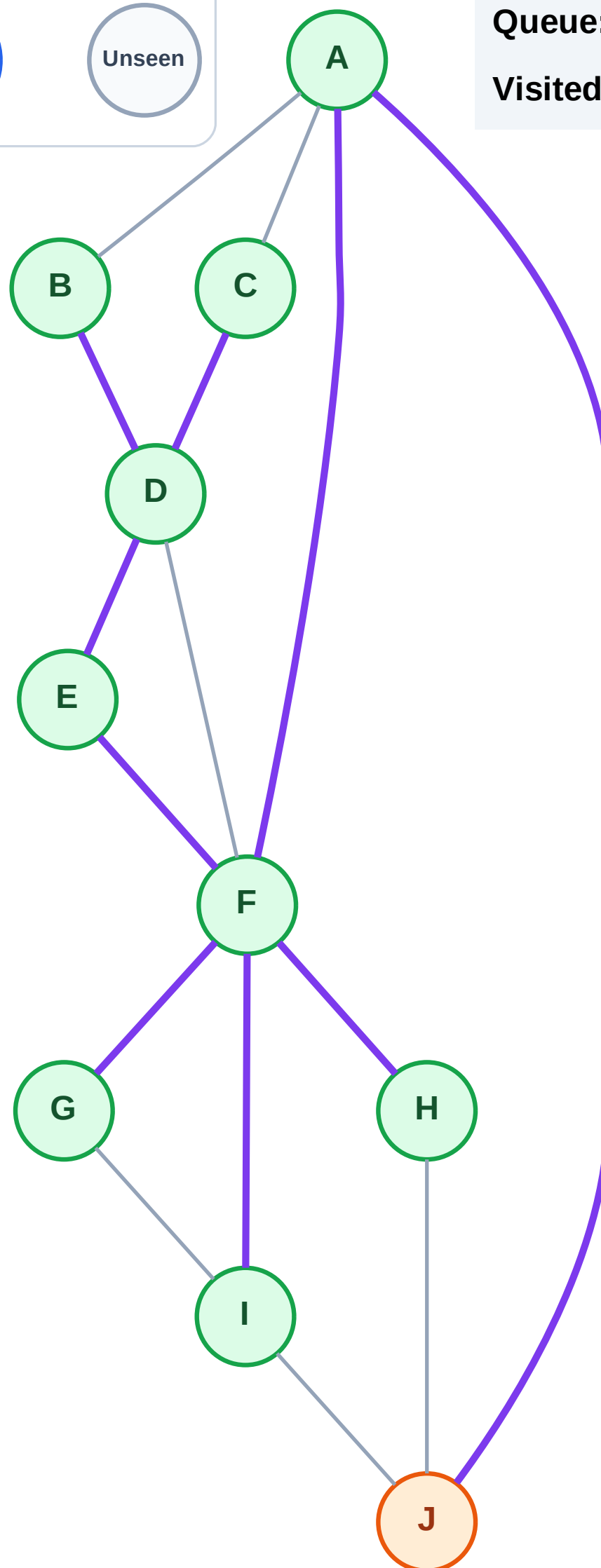
Step 20: Process node J

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

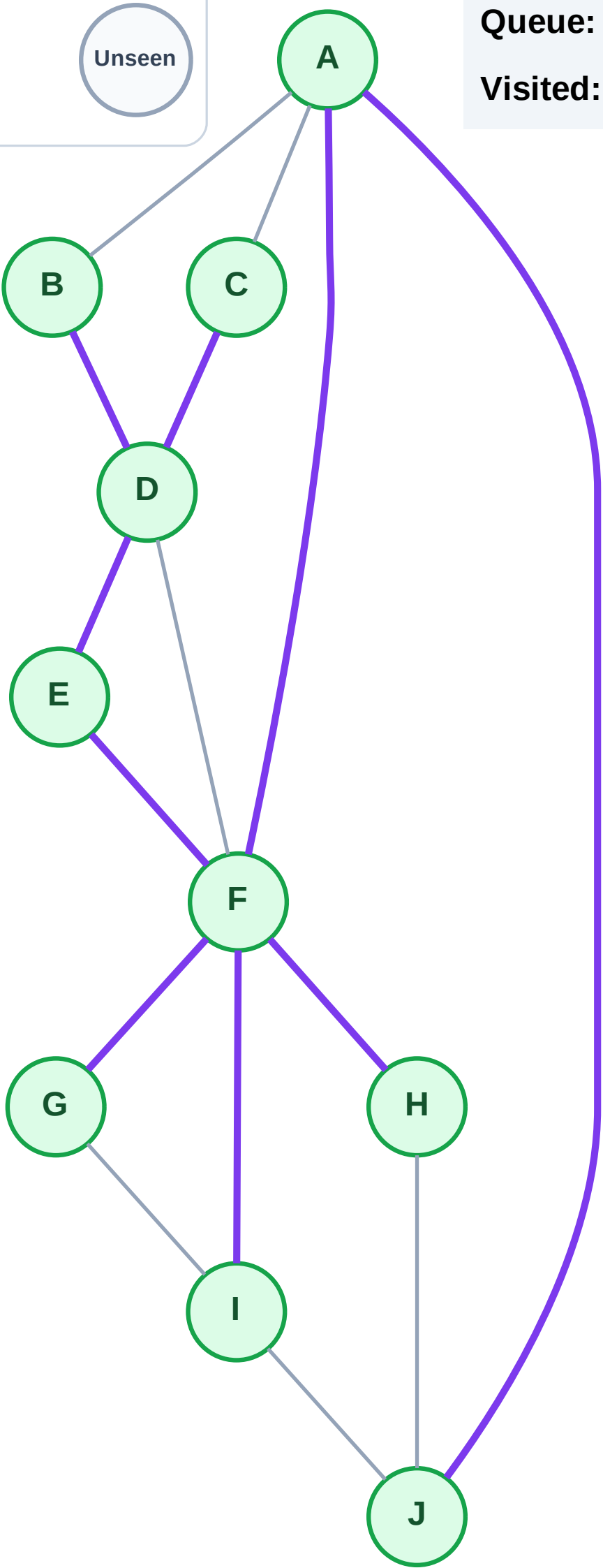
Step 21: No new neighbors from J

Legend



Queue: (empty)

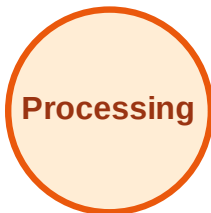
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

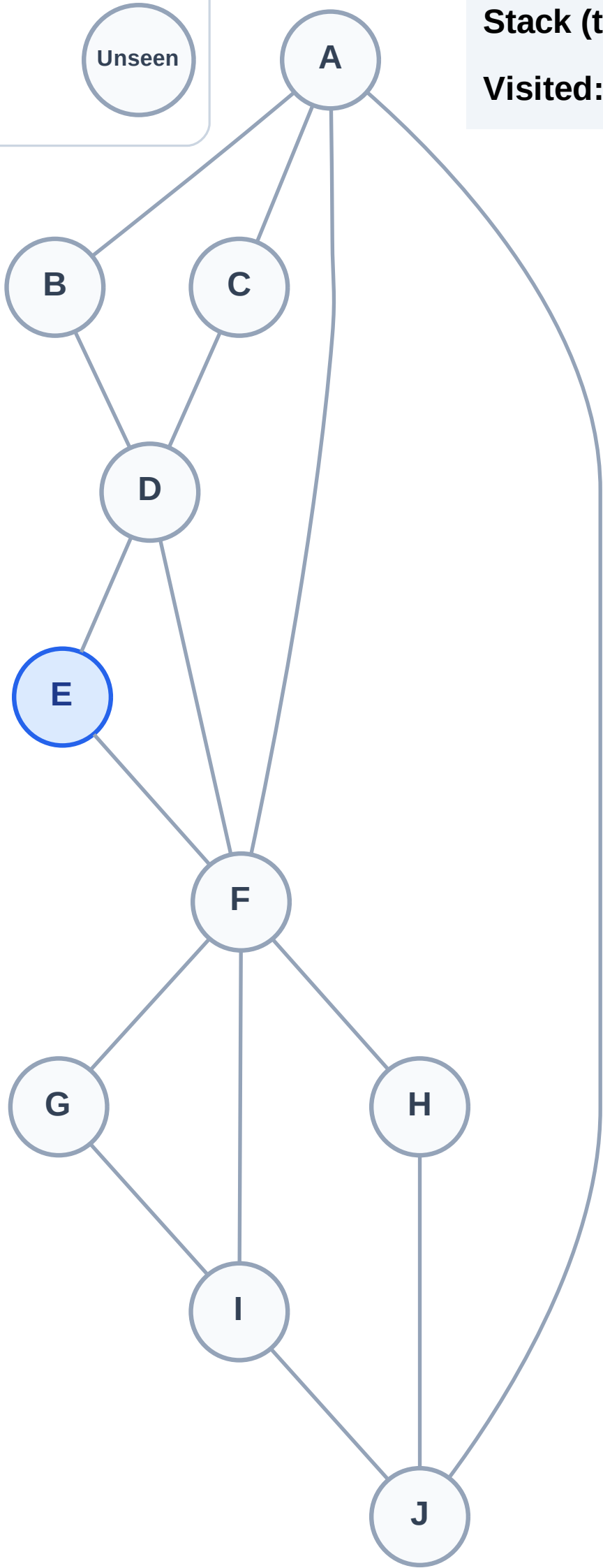
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

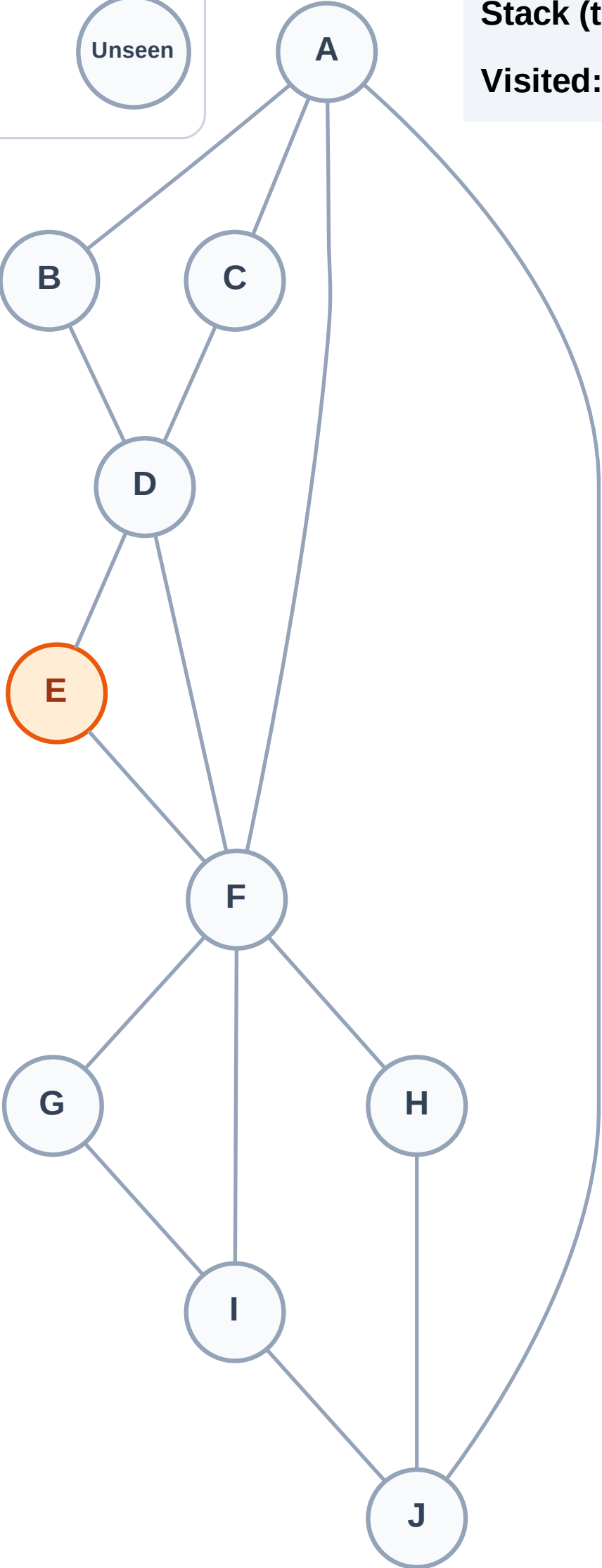
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

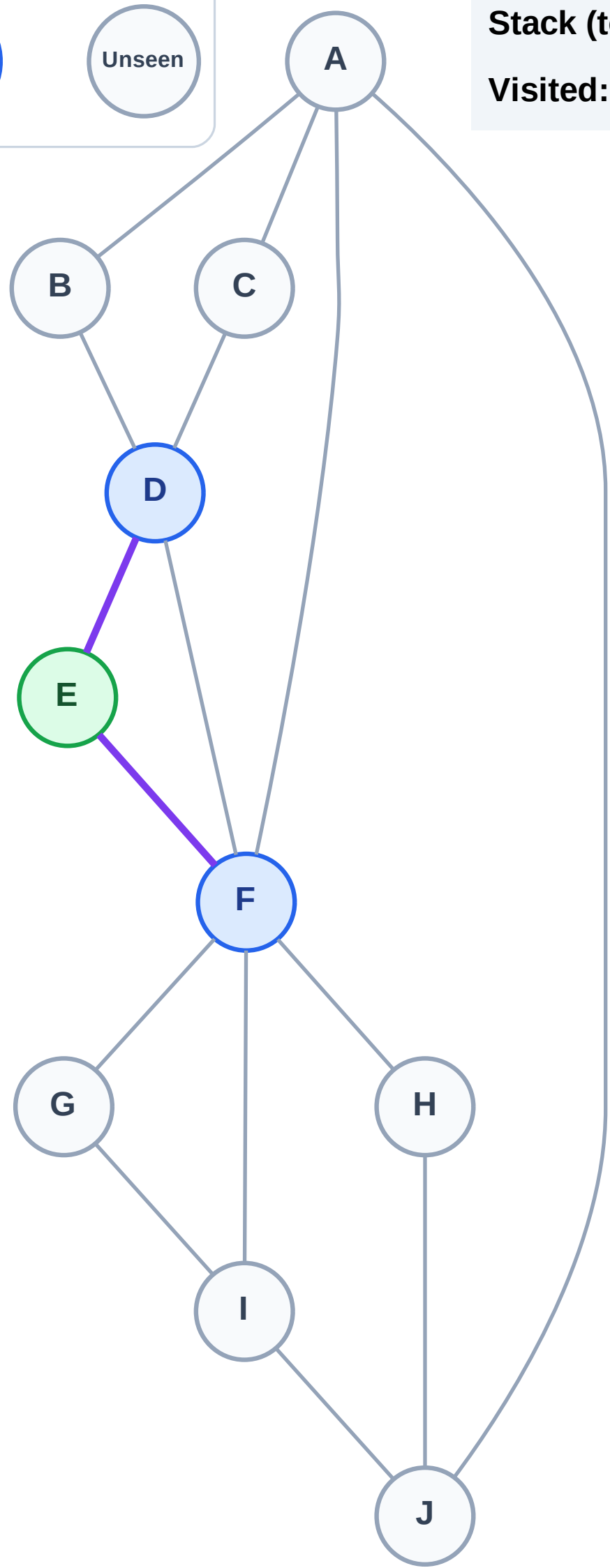
Step 3: Discover F, D from E

Legend



Stack (top at right): F | D

Visited: E



DFS Traversal

Step 4: Process node D

Legend

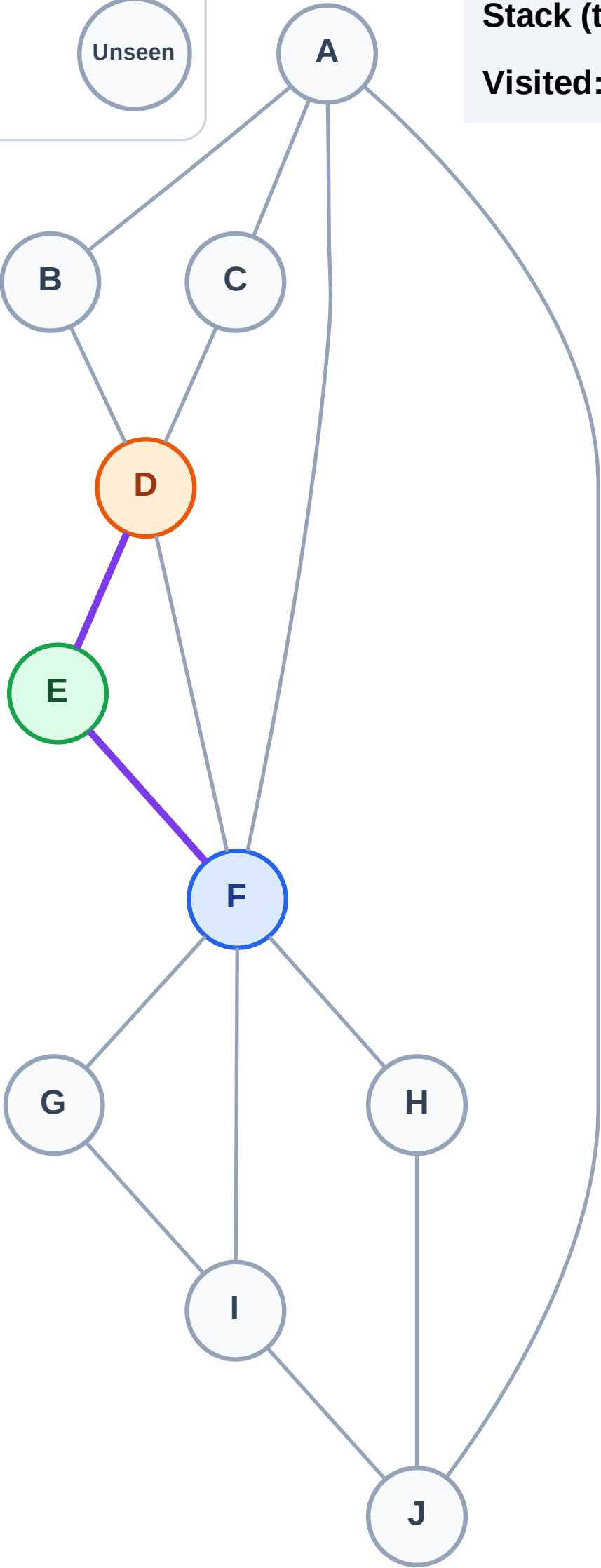
Complete

Processing

Discovered

Unseen

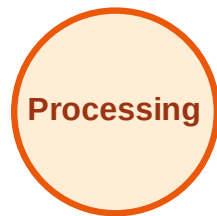
Stack (top at right): F
Visited: E



DFS Traversal

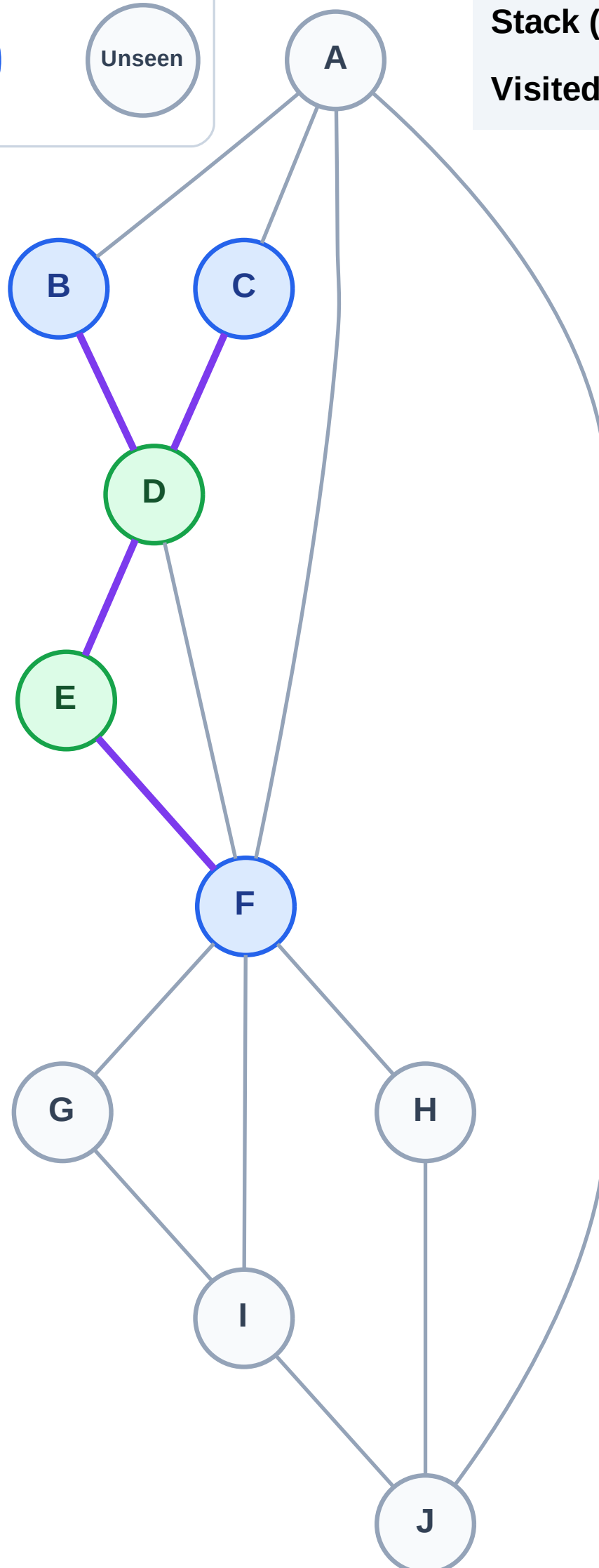
Step 5: Discover C, B from D

Legend



Stack (top at right): F | C | B

Visited: D, E



DFS Traversal

Step 6: Process node B

Legend

Complete

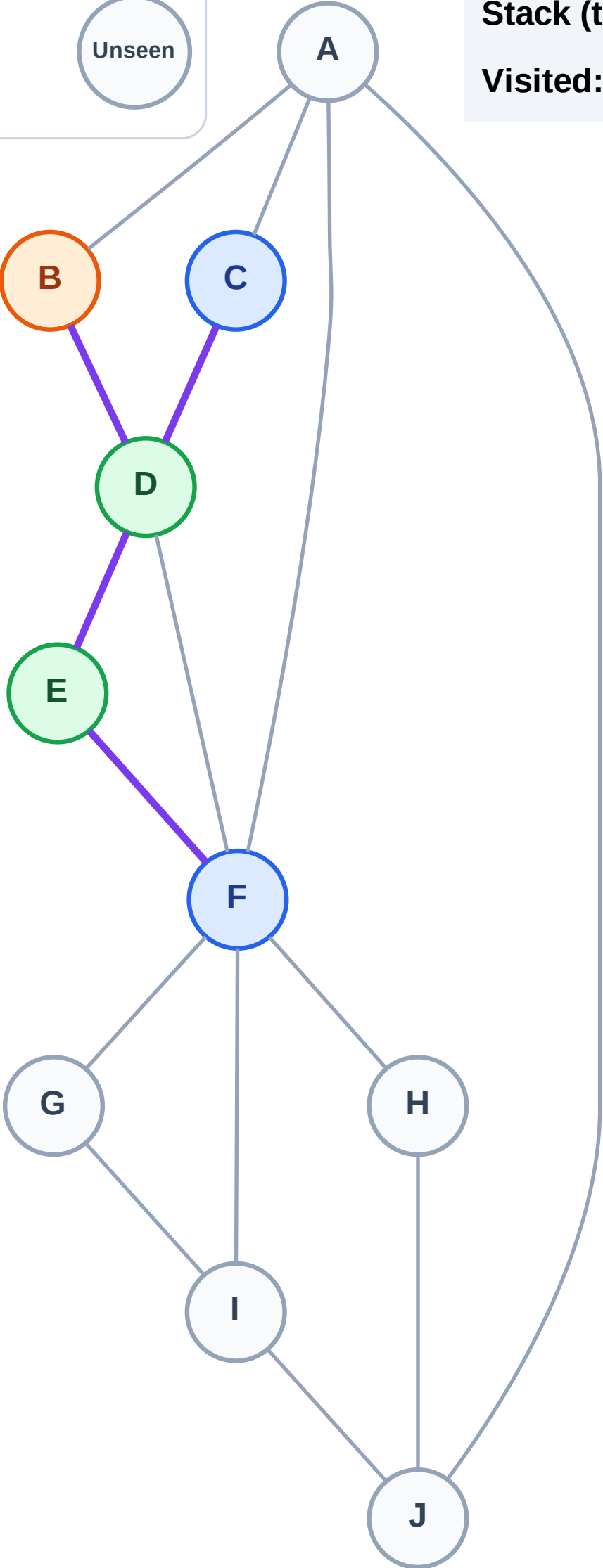
Processing

Discovered

Unseen

Stack (top at right): F | C

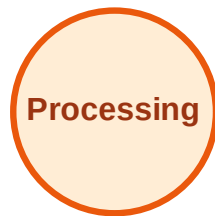
Visited: D, E



DFS Traversal

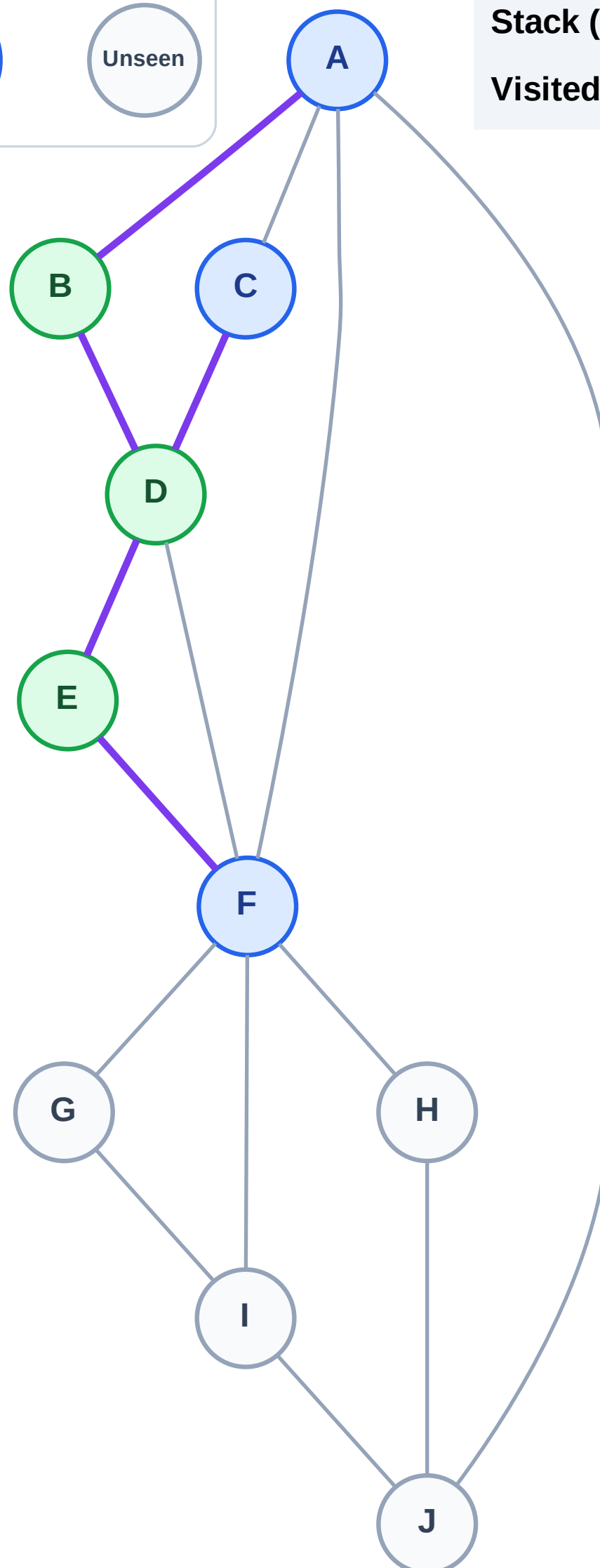
Step 7: Discover A from B

Legend



Stack (top at right): F | C | A

Visited: B, D, E



DFS Traversal

Step 8: Process node A

Legend

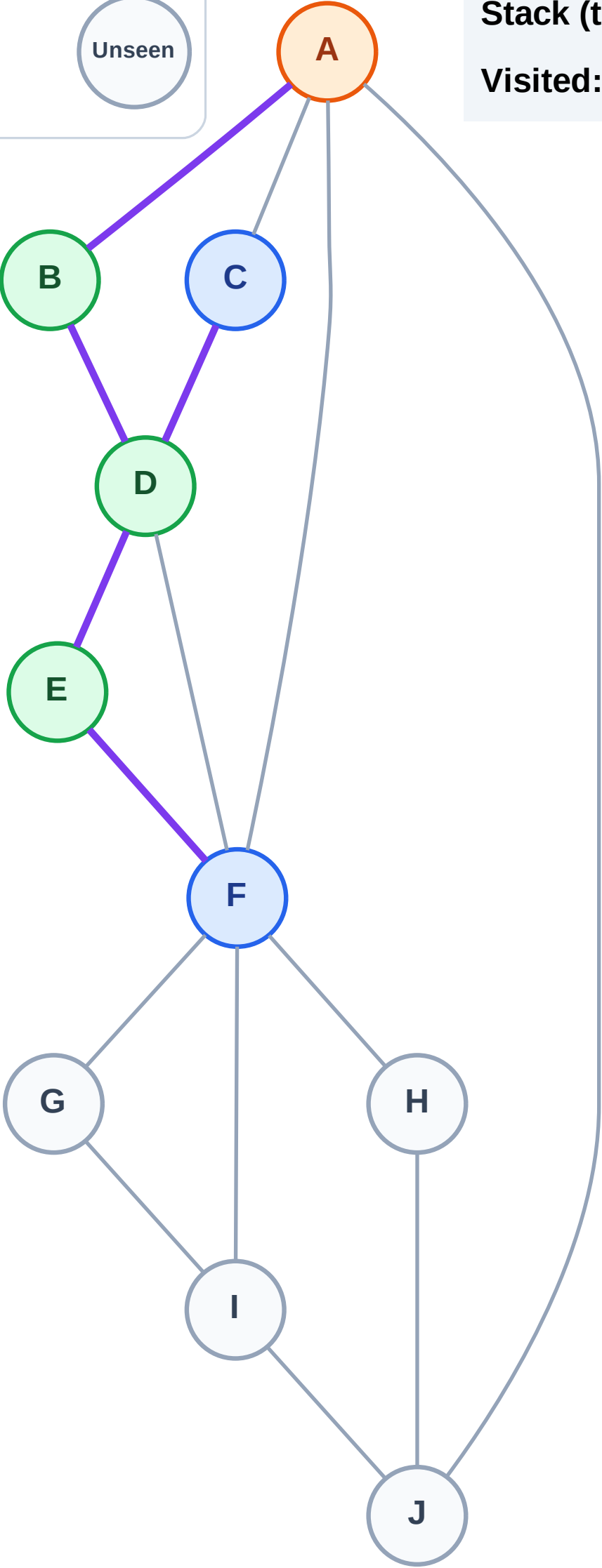
Complete

Processing

Discovered

Unseen

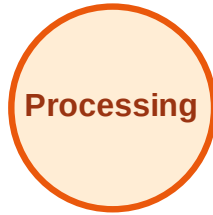
Stack (top at right): F | C
Visited: B, D, E



DFS Traversal

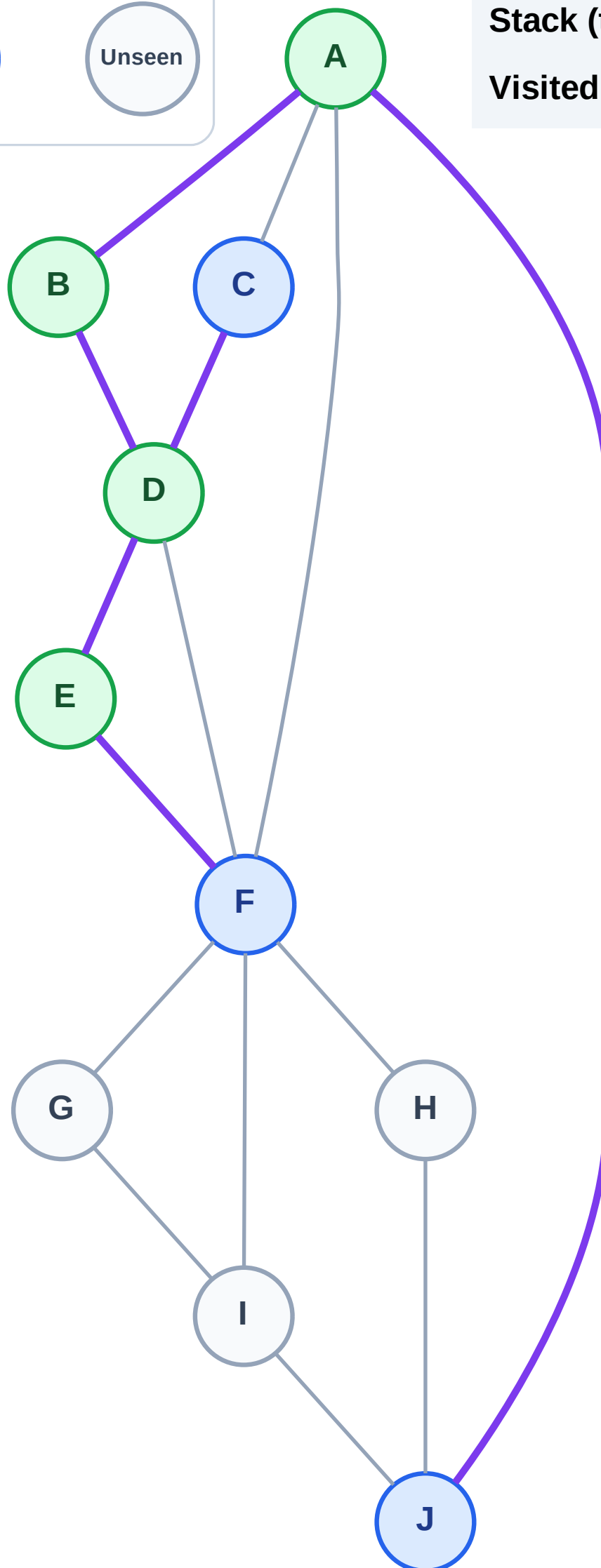
Step 9: Discover J from A

Legend



Stack (top at right): F | C | J

Visited: A, B, D, E



DFS Traversal

Step 10: Process node J

Legend

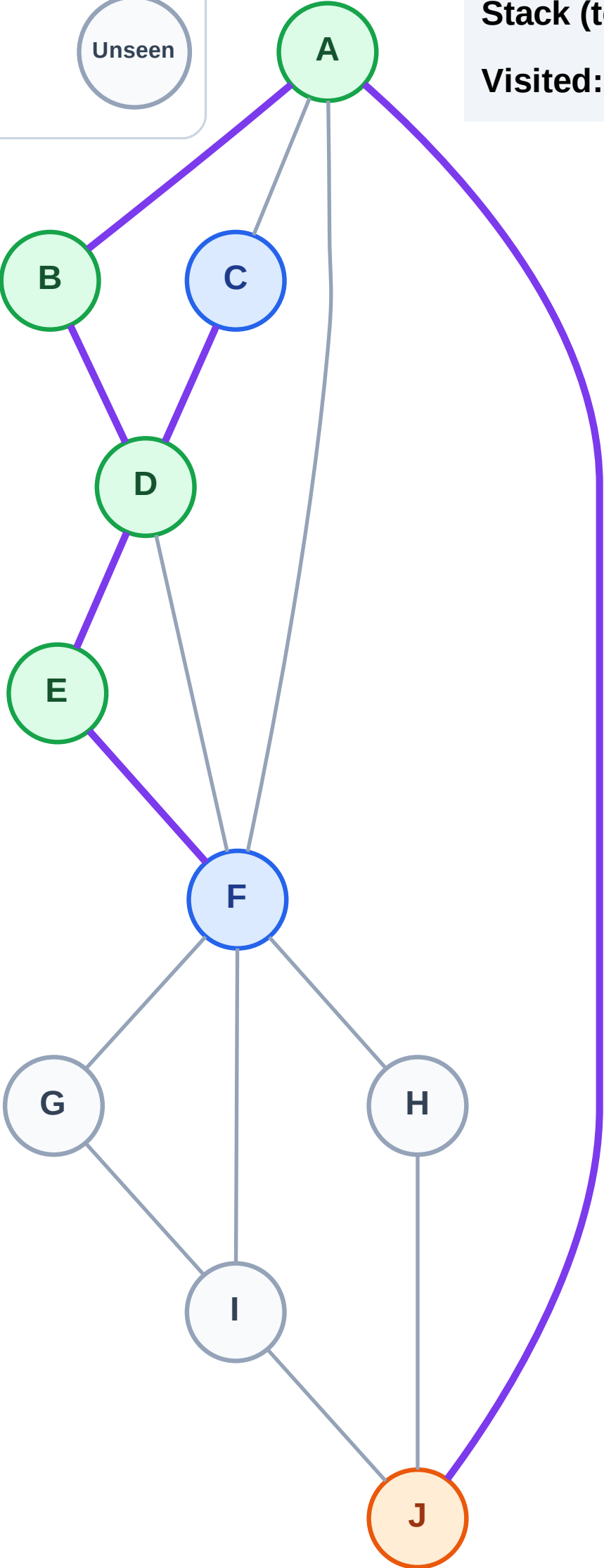
Complete

Processing

Discovered

Unseen

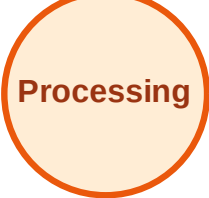
Stack (top at right): F | C
Visited: A, B, D, E



DFS Traversal

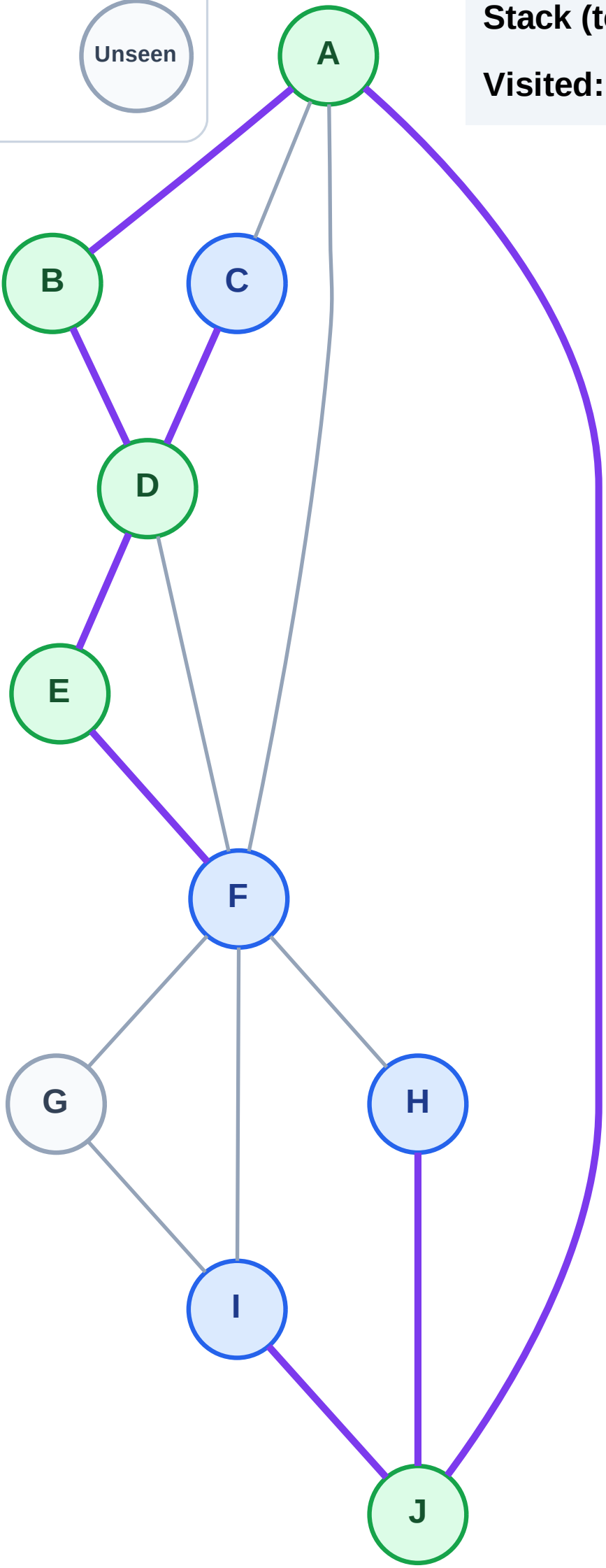
Step 11: Discover I, H from J

Legend



Stack (top at right): F | C | I | H

Visited: A, B, D, E, J



DFS Traversal

Step 12: Process node H

Legend

Complete

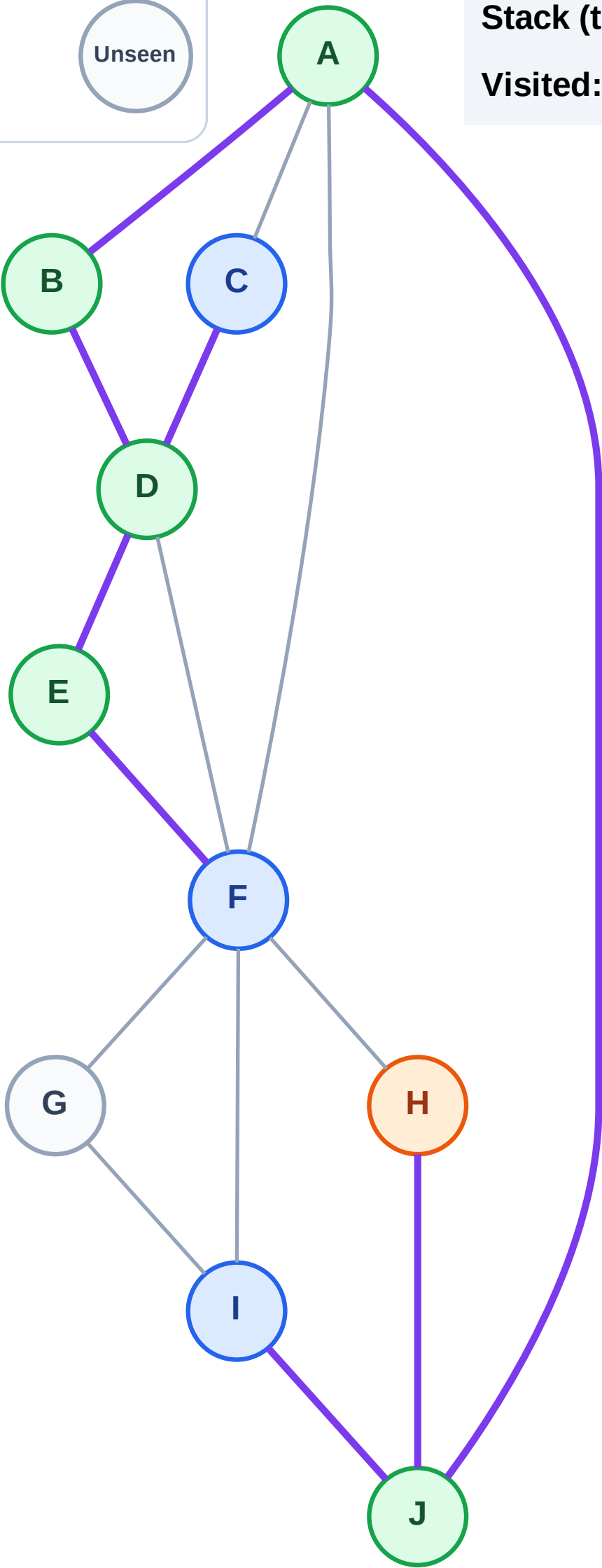
Processing

Discovered

Unseen

Stack (top at right): F | C | I

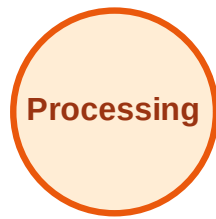
Visited: A, B, D, E, J



DFS Traversal

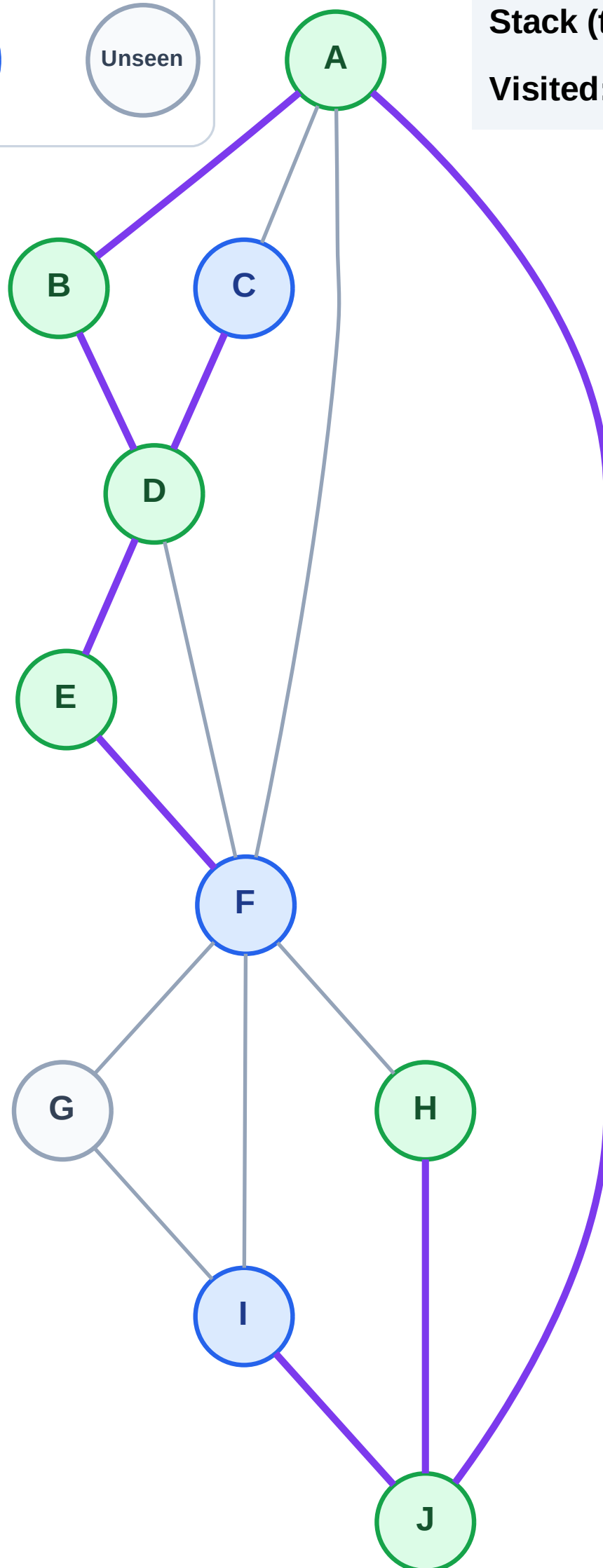
Step 13: No new neighbors from H

Legend



Stack (top at right): F | C | I

Visited: A, B, D, E, H, J



DFS Traversal

Step 14: Process node I

Legend

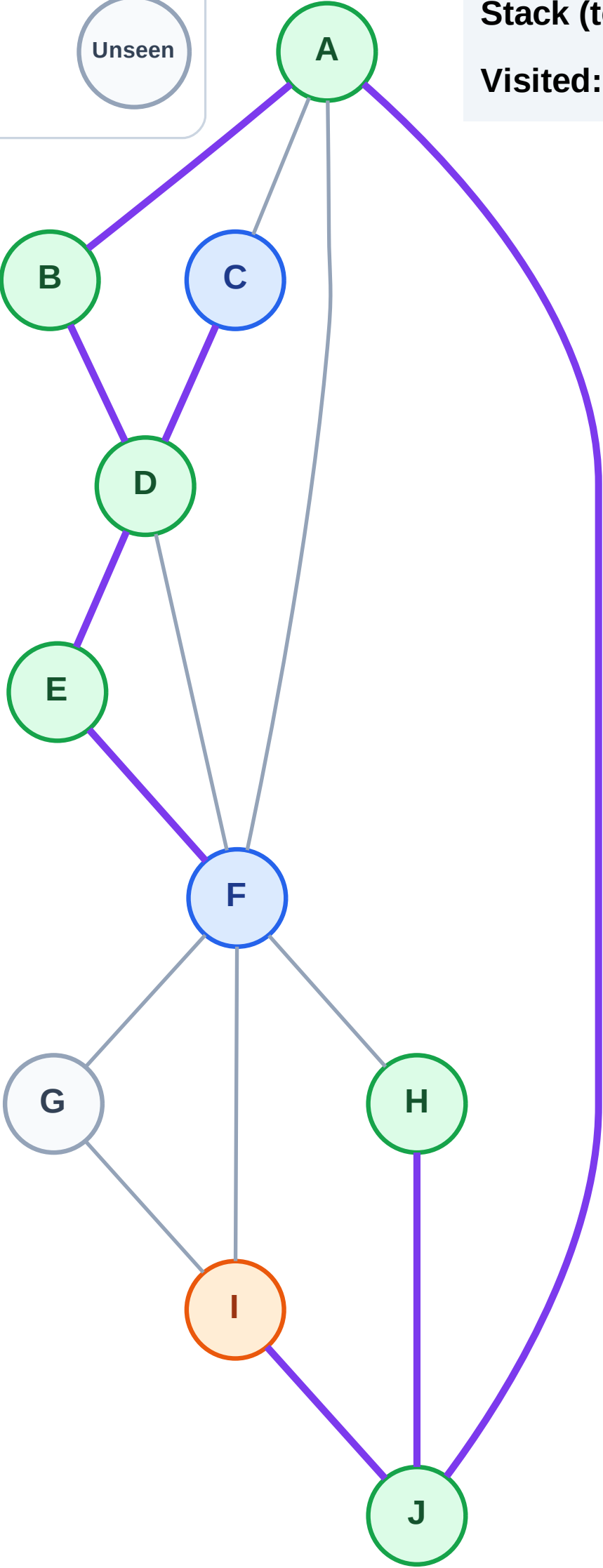
Complete

Processing

Discovered

Unseen

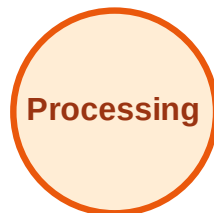
Stack (top at right): F | C
Visited: A, B, D, E, H, J



DFS Traversal

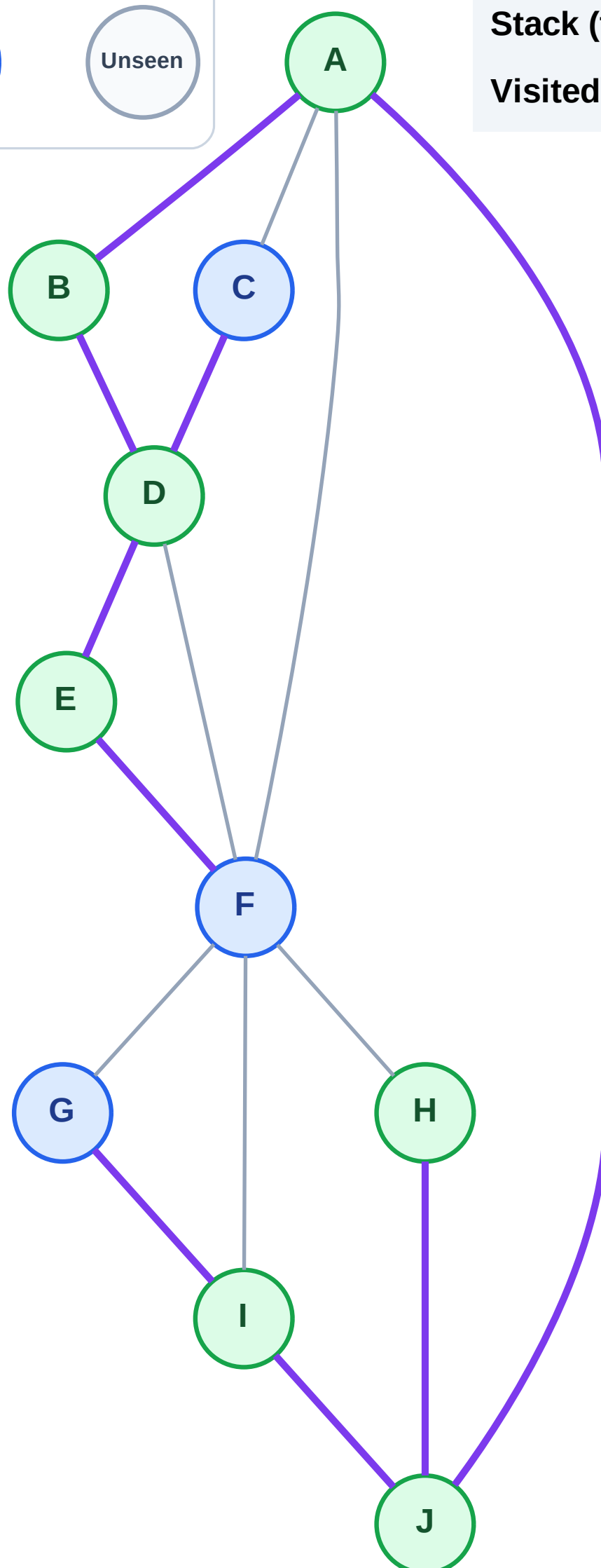
Step 15: Discover G from I

Legend



Stack (top at right): F | C | G

Visited: A, B, D, E, H, I, J



DFS Traversal

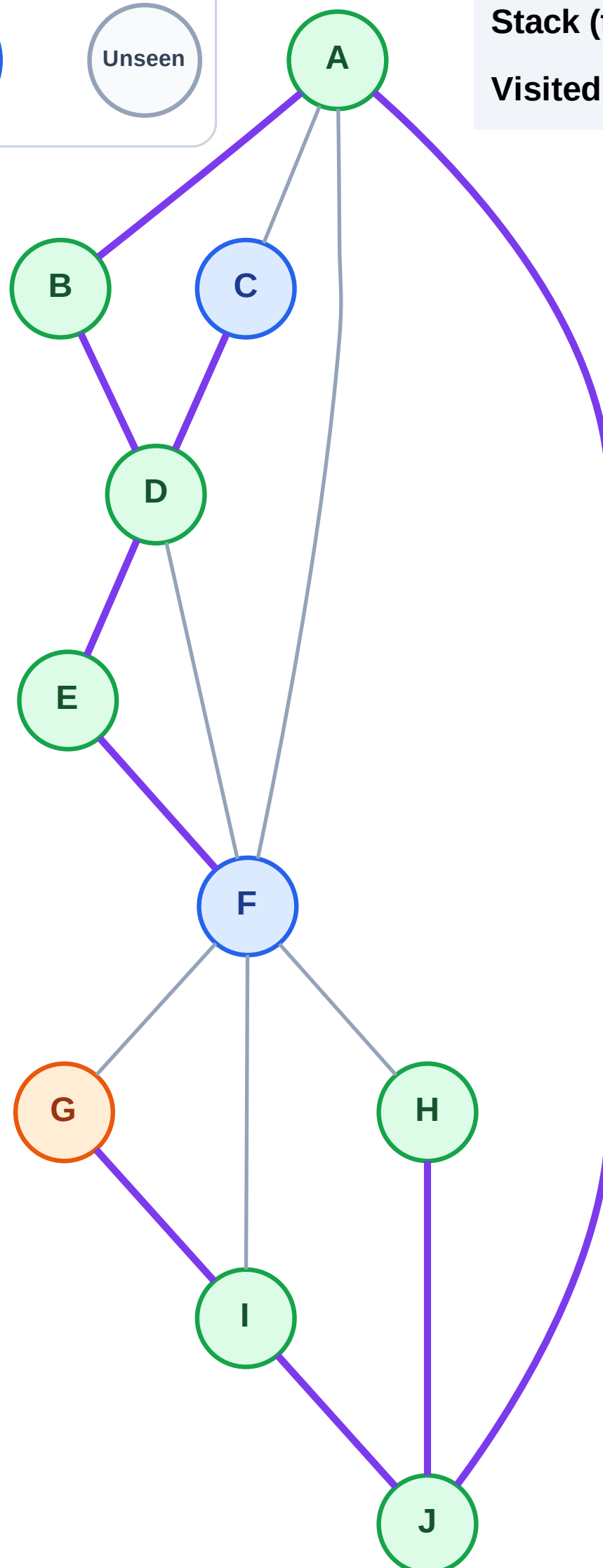
Step 16: Process node G

Legend



Stack (top at right): F | C

Visited: A, B, D, E, H, I, J



DFS Traversal

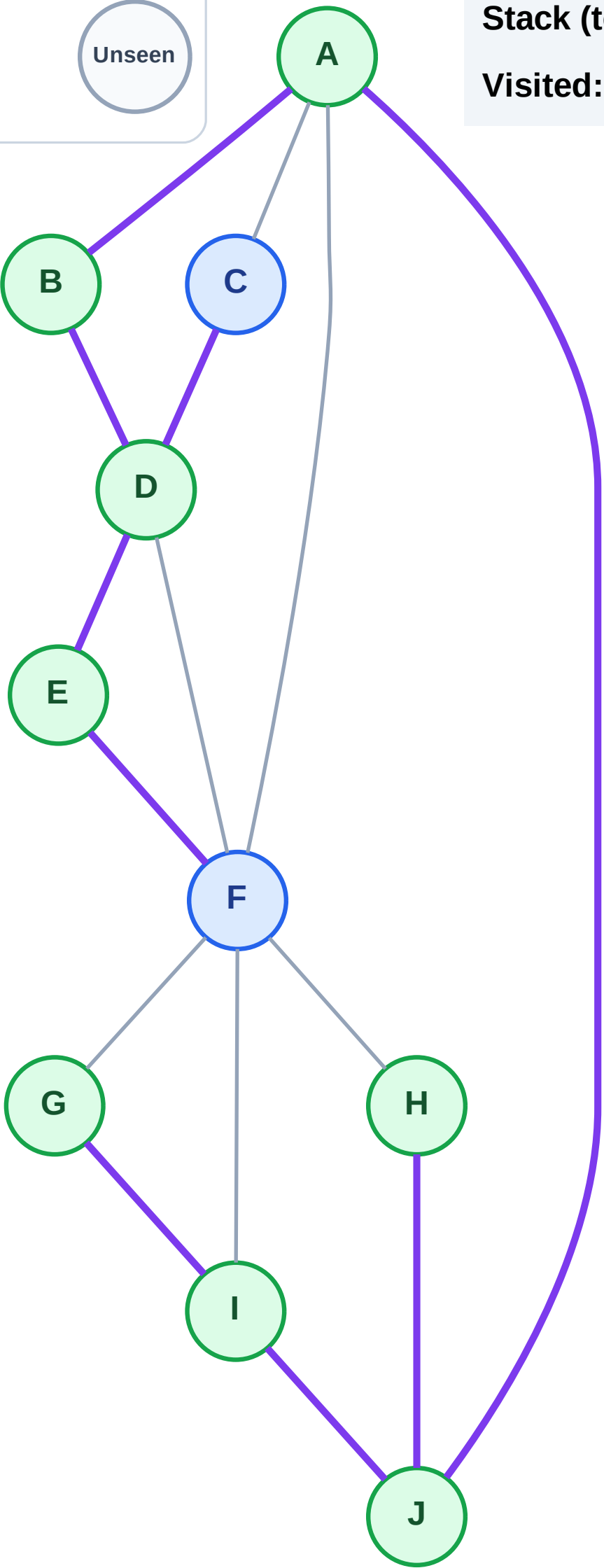
Step 17: No new neighbors from G

Legend



Stack (top at right): F | C

Visited: A, B, D, E, G, H, I, J



DFS Traversal

Step 18: Process node C

Legend

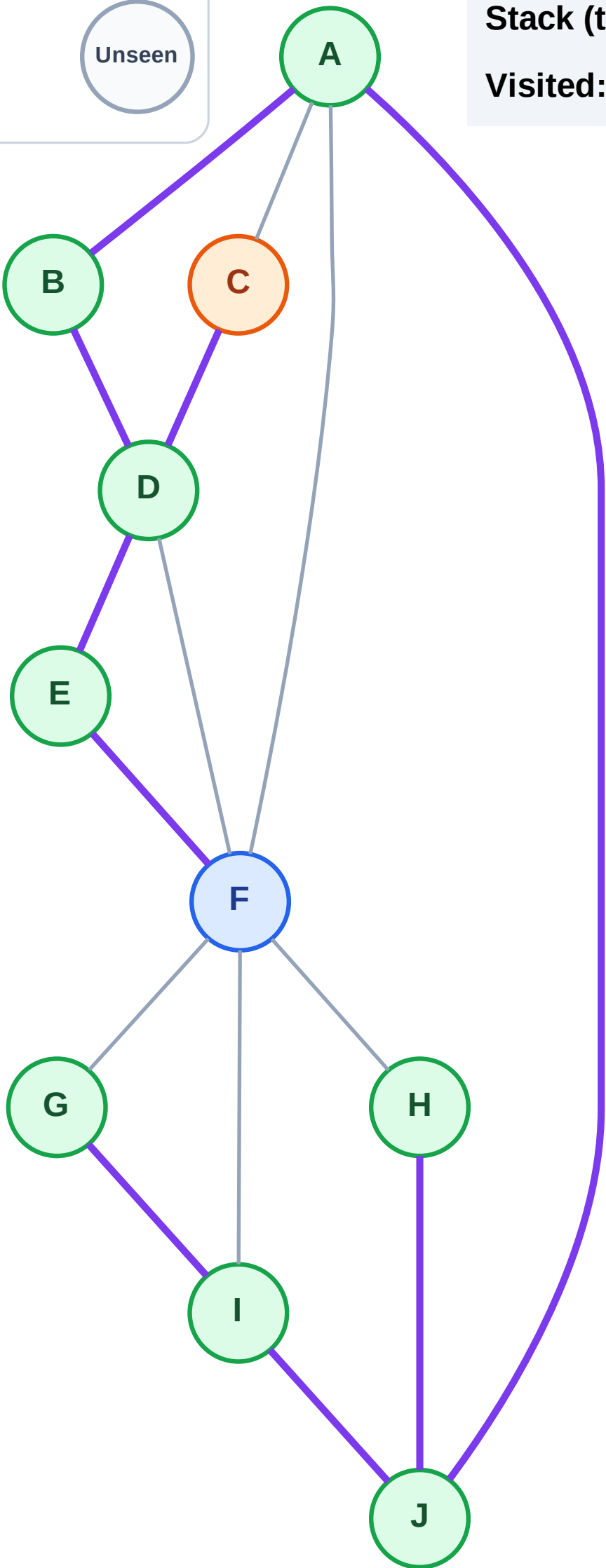
Complete

Processing

Discovered

Unseen

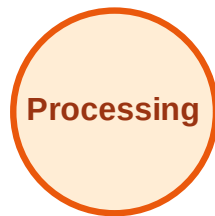
Stack (top at right): F
Visited: A, B, D, E, G, H, I, J



DFS Traversal

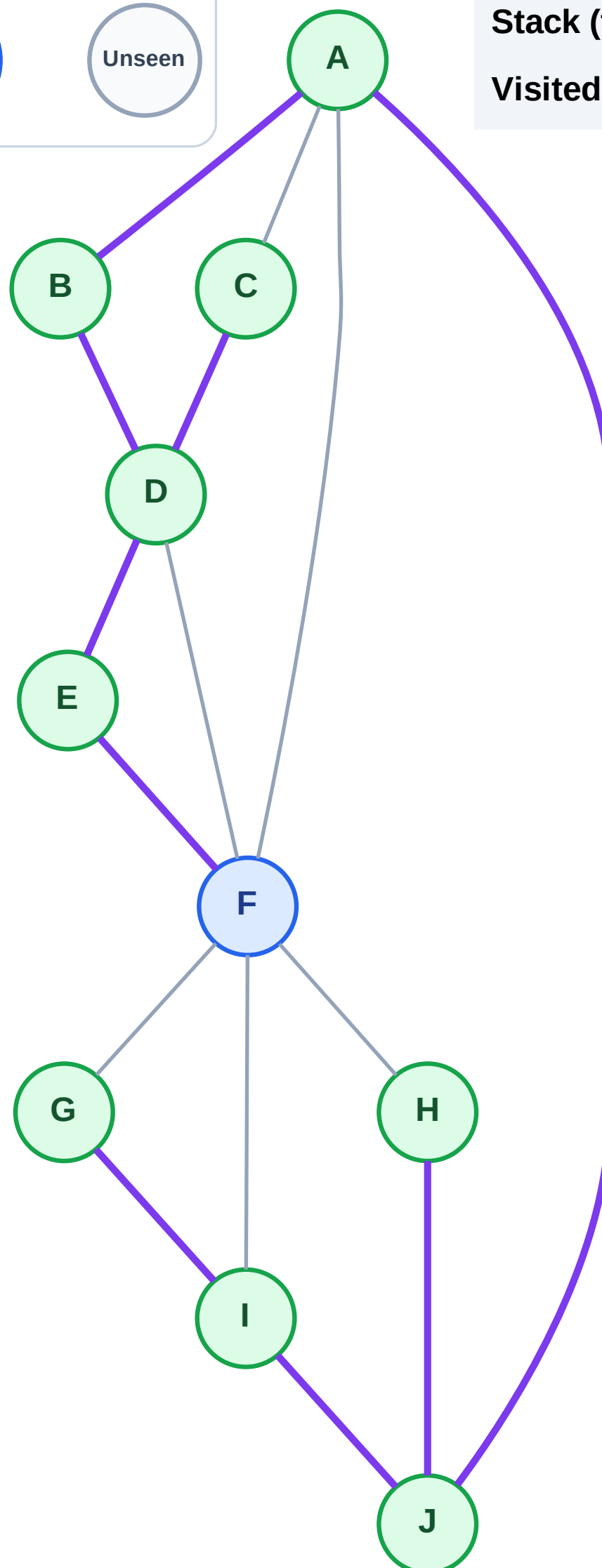
Step 19: No new neighbors from C

Legend



Stack (top at right): F

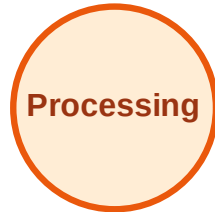
Visited: A, B, C, D, E, G, H, I, J



DFS Traversal

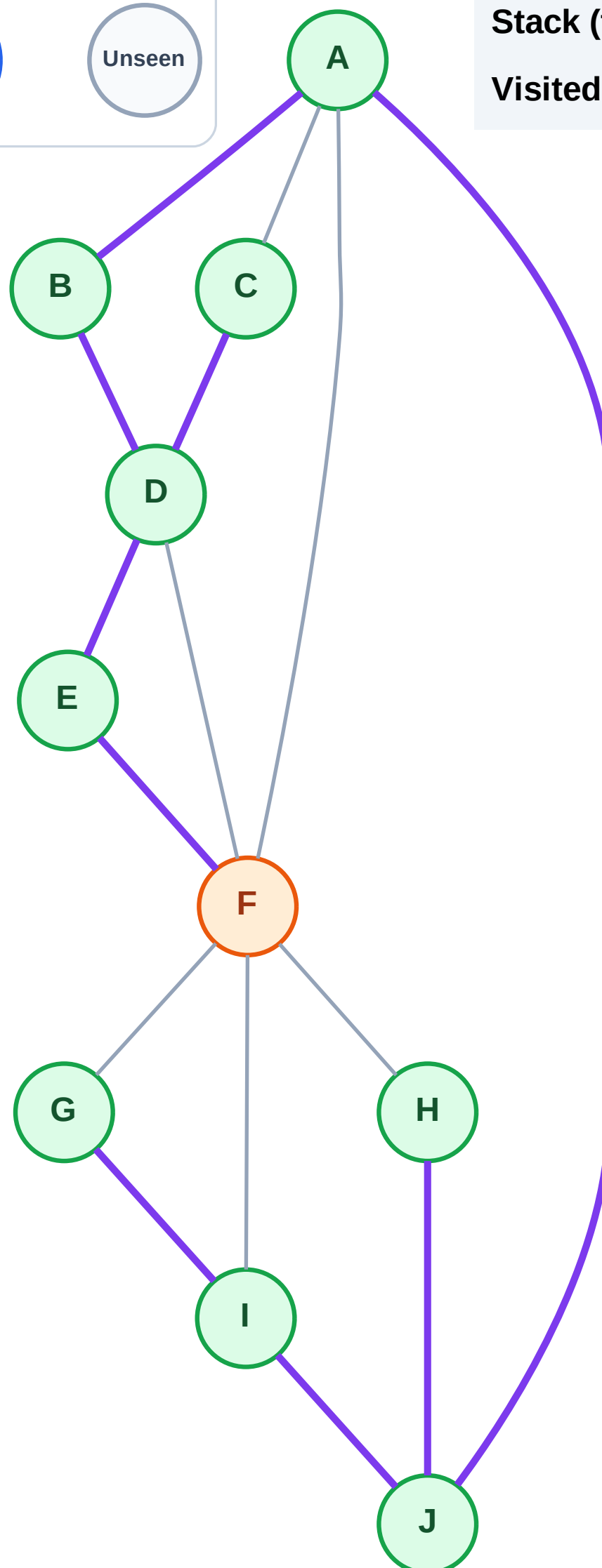
Step 20: Process node F

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, G, H, I, J



DFS Traversal

Step 21: No new neighbors from F

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

